



A survey of deep learning for industrial visual anomaly detection

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Abstract

Industrial visual anomaly detection is critical for ensuring system reliability, safety, and efficiency. This paper presents a comprehensive survey of state-of-the-art anomaly detection techniques, analyzing methodologies, implementations, and recent advancements. Our survey aims to accelerate researchers' understanding of emerging trends while providing a structured foundation for newcomers. We systematically review 196 recent papers covering five learning strategies, including fully supervised, semi-supervised, self-supervised, weakly supervised, and unsupervised approaches. This paper provides a detailed introduction to twelve industrial anomaly detection methods, revealing their theoretical foundations, technical principles, and practical applications. Additionally, we provide a detailed overview to 2D and 3D datasets for industrial visual anomaly detection. In addition, we critically analyze and summarize the experimental results, identify key performance indicators, and discuss the latest trends in the field of industrial anomaly detection. Beyond analysis, we contribute actionable insights for selecting optimal models for real-world deployment. Finally, we highlight open challenges and outline future research directions to drive innovation in this evolving field. The detailed resources are available at <https://github.com/IHPCRits/IAD-Survey>.

Keywords Deep learning · Computer vision · Industrial visual anomaly detection

1 Introduction

Industrial visual anomaly detection is a critical technique for ensuring product quality and improving production efficiency in modern manufacturing. With the advancement of computer vision and deep learning technologies, the field is rapidly transitioning from traditional automation to intelligent visual inspection systems. Industrial visual anomaly detection dif-

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fers fundamentally from general visual anomaly detection primarily due to the distinct challenges posed by industrial environments. Specifically, industrial visual anomaly detection often involves anomalies characterized by subtle appearance variations, limited availability of defective samples, and stringent requirements on detection accuracy and real-time processing. In contrast, general visual anomaly detection typically deals with more diverse anomaly types without the stringent constraints of industrial applications. Moreover, anomalies in industrial settings can frequently exhibit specific structural and semantic characteristics, such as minor surface defects, precise dimensional deviations, or logical misalignments that are challenging to capture using standard visual anomaly detection approaches. Industrial visual anomaly detection faces several practical challenges inherent to real-world industrial settings. First, the scarcity or complete absence of labeled anomaly data presents a major obstacle, complicating supervised learning approaches and demanding the development of effective unsupervised or semi-supervised techniques. Second, variability in manufacturing conditions—including changes in lighting, camera angles, and environmental factors—can significantly affect the visual appearance of products, requiring anomaly detection methods to maintain robustness under diverse operating conditions. Third, industrial datasets often exhibit severe imbalance, as normal instances vastly outnumber anomalous ones, leading to challenges in accurately modeling anomalies without incurring excessive false positives. Lastly, high levels of image noise due to sensor imperfections or harsh environmental conditions further compound detection difficulty. As production processes become increasingly refined and production lines operate at higher speeds, manual inspection can no longer meet the dual demands of efficiency and accuracy. This has led to a growing need for high-precision, automated, and adaptive visual anomaly detection techniques. In recent years, deep learning-based approaches have emerged as the dominant paradigm in this field, achieving remarkable performance across various benchmarks. However, the diversity of data types, detection tasks, and model architectures has resulted in a large volume of fragmented research. Despite this progress, there remains a lack of systematic surveys that comprehensively organize, categorize, and compare existing methods. Based on this, we provide a structured and in-depth survey of recent deep learning-based methods for industrial visual anomaly detection.

Recently, Li et al. (2023) provide a comprehensive and structured survey of state-of-the-art explainable anomaly detection techniques, presenting a taxonomy based on characterizing the key aspects of each explainable anomaly detection technique, with the aim of helping practitioners and researchers to find the explainable anomaly detection method that best suits their needs. Cao et al. (2024) investigate a survey of the challenges, methods, and prospects for visual anomaly detection, in which they investigate visual anomaly detection by outlining its background and conceptual definitions while systematically categorizing, emphasizing, and discussing recent advances from the perspectives of sample size, data modality, and anomaly hierarchy. Liu et al. (2024) present a comprehensive review of deep learning-based image anomaly detection techniques from the perspectives of neural network architecture, supervision level, loss function, metrics, and datasets. Lin et al. (2024) summarize a review of RGB, 3D and Multimodal methods for unsupervised industrial anomaly detection, providing a comprehensive summary of unsupervised industrial anomaly detection in all three modal environments. While these surveys offer valuable insights, they either focus on general image anomaly detection, specific learning paradigms, or individual data modalities. To bridge this gap and offer a more complete perspective tailored to real-

world industrial applications, we conduct a comprehensive survey on deep learning-based industrial visual anomaly detection. Specifically, we conducted an in-depth study of deep learning-based industrial vision anomaly detection methods and applications, covering five learning paradigms: fully supervised, semi-supervised, weakly supervised, self-supervised, and unsupervised learning. We systematically discussed solutions for scenarios involving scarce defect samples or noisy labels. Beyond covering major deep learning paradigms, we compile and analyze open-source resources, including widely used datasets, performance metrics, publicly available codebases, and benchmark comparisons. This facilitates reproducibility, enhances practical relevance, and supports more efficient research. Furthermore, we propose a structured taxonomy of industrial visual anomaly detection methods based on their learning strategies, aligning them with real-world annotation conditions and deployment constraints. Throughout this survey, we highlight impactful work in industrial visual anomaly detection. This study aims to comprehensively analyze the latest developments in industrial visual anomaly detection, covering two detection granularities: image-level (classifying the entire image as normal or abnormal) and pixel-level (locating abnormal regions). Additionally, we investigate industrial visual anomaly detection methods under multi-class anomaly settings. The contributions of this study are summarized as follows:

- We systematically organize the existing research paradigms based on learning strategies and supervision levels, providing precise methodological guidance according to different types of annotation availability.
- We present a comprehensive analysis of industrial anomaly detection methods, highlighting their applicability and limitations in real-world production environments.
- We summarize key challenges and research gaps in this field and outline promising future research directions to support further development.

Figure 1 illustrates the overall organization of this paper. Section 2 provides an in-depth and structured overview of deep learning-based industrial anomaly detection, covering both learning approaches and implementation techniques. Section 3 summarizes the industrial visual anomaly detection datasets. Section 4 presents the evaluation metrics and experimental results in industrial visual anomaly detection. Section 5 gives an outlook on the challenge and future directions in industrial visual anomaly detection. Finally, we conclude this survey in Sect. 6.

2 Deep learning for industrial visual anomaly detection

To provide a structured and coherent overview of industrial anomaly detection methods, we categorize the literature from two complementary perspectives: learning strategy and methodological implementation. The first perspective emphasizes the degree of supervision, including fully supervised, semi-supervised, weakly supervised, self-supervised, and unsupervised learning. These categories reflect the availability of labeled data in different industrial scenarios. The second perspective focuses on the core methodological characteristics such as reconstruction-based models, feature embedding, generative modeling, and detection strategies. In practice, many methods lie at the intersection of these two dimensions. For instance, an unsupervised method may employ a feature embedding mechanism

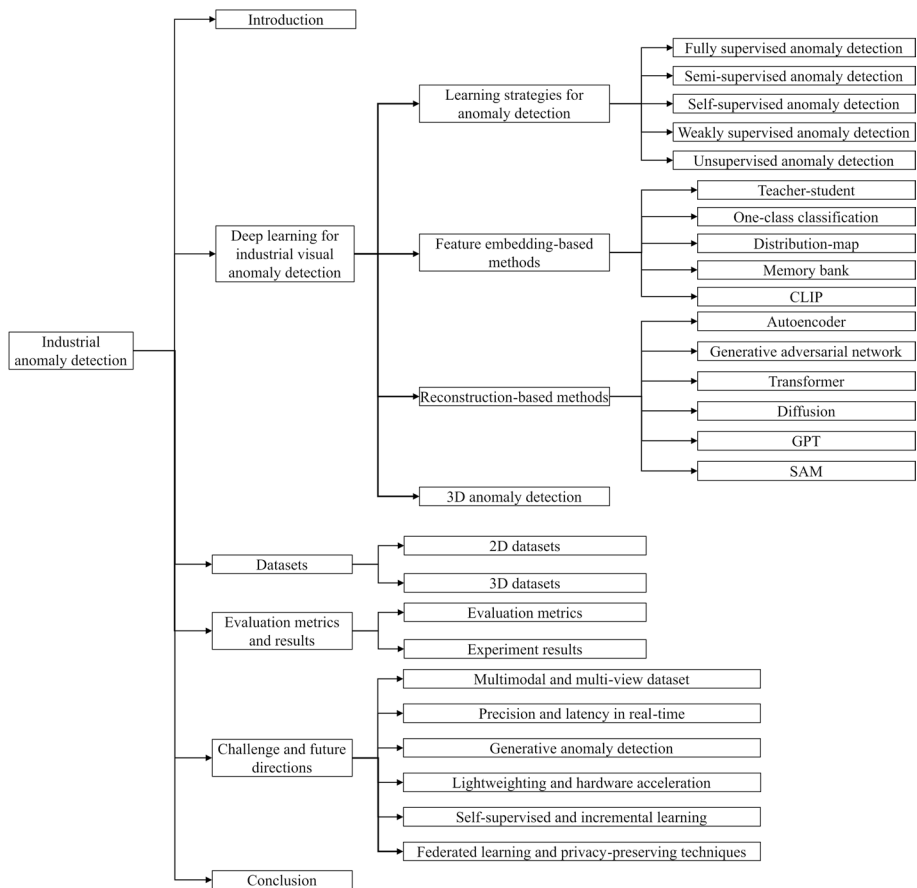


Fig. 1 Organization of this survey on industrial anomaly detection, illustrating the main topics (learning strategies, methodology, datasets, and future directions) covered in each section

or leverage a generative model such as GAN or diffusion. By adopting this dual-axis framework, this survey aims to present a more complete landscape of industrial visual anomaly detection techniques, while providing researchers with multiple pathways for understanding and selecting appropriate approaches.

2.1 Learning strategies for anomaly detection

In different learning paradigms for industrial vision anomaly detection, the composition of annotated data varies significantly. Fully supervised methods rely on complete and detailed annotations-i.e., pixel-level or instance-level labels for normal samples and all known defect categories. These methods incur the highest annotation costs but provide the most accurate semantic information about defects. In semi-supervised scenarios, only a small portion of normal and abnormal images are fully labeled, while the large-scale unlabeled data contains both normal images and images with unknown defects. These models balance annotation cost and performance by leveraging a small amount of high-quality labeled data together

with a large volume of raw data. Self-supervised methods apply a uniform “normal” label only to normal images and do not require any defect annotations. When needed, additional unlabeled normal images can be introduced for pretext tasks to learn richer feature representations. In contrast, weakly supervised methods use only coarse-grained, image-level OK/NG labels for normal and abnormal samples. They do not involve pixel-level or instance-level annotations, which significantly reduces manual labeling costs but also introduces issues such as low localization accuracy and label noise. Finally, unsupervised methods rely entirely on annotations from normal samples, where all normal images are labeled simply as “normal,” with no defect information. The model learns to detect anomalies by identifying deviations from the learned normal distribution, though this approach introduces greater challenges in threshold determination and false positive control. Table 1 summarizes the advantages, limitations, and application scenarios of various supervised learning methods for industrial visual anomaly detection.

This subsection provides an overview of the supervised learning methods used in industrial visual anomaly detection. Section 2.1.1 introduces fully supervised learning methods. Section 2.1.2 presents semi-supervised learning methods. Section 2.1.3 discusses self-supervised learning methods. Section 2.1.4 details weakly supervised learning methods. Section 2.1.5 explores unsupervised learning methods.

Table 1 Comparison of industrial anomaly detection methods under different learning strategies

Learning strategy	Advantages	Limitations	Typical application scenarios
Fully supervised anomaly detection	High precision, Clear labels	High labeling costs and difficulty in responding to new anomalies	Known anomaly types high precision
Semi-supervised anomaly detection	Combining a small amount of labeled data and a large amount of unlabeled data improves the detection effect	The balance between labeled and unlabeled data is complex, and performance depends on the quality of the unlabeled data	Samples of labeling are scarce
Self-supervised anomaly detection	Combining a small amount of labeled data and a large amount of unlabeled data improves the detection effect	The balance between labeled and unlabeled data is complex, and performance depends on the quality of the unlabeled data	Samples of labeling are scarce
Weakly supervised anomaly detection	Reduce labeling costs by using low-quality or incomplete labeling data	Low-quality labeling affects model performance, and needs to deal with label noise	The sample labeling is inaccurate
Unsupervised anomaly detection	Without labeling data, unknown anomalies are automatically identified	The error rate is high, making it difficult to accurately define an anomaly	High labeling costs

2.1.1 Fully supervised anomaly detection

Although fully supervised anomaly detection has advantages in terms of detection accuracy, its reliance on large amounts of labeled data and limited generalization ability constrain its practical applications. Furthermore, industrial defects typically follow a long-tailed distribution, with a very small number of critical defects accounting for only a tiny fraction of the total samples. To address these challenges, recent research has focused on reducing labeling costs, improving model generalization, and enabling real-time detection—thereby enhancing the practicality and effectiveness of fully supervised anomaly detection. Meanwhile, to mitigate data imbalance, current approaches generally tackle the problem from two perspectives: loss function reweighting (e.g., Focal Loss, Class-Balanced Loss) and sample-level resampling. The former increases the gradient contribution of hard examples or minority classes to help the model focus on them more effectively; the latter leverages techniques such as hard example mining, class-balanced batch sampling, and minority class oversampling—including synthetic defect generation via GANs or diffusion models—to increase the effective exposure of rare defects. Currently, YOLO-based methods such as YOLOv3-Tiny, YOLOv5, and YOLOv8 are widely used in fully supervised industrial anomaly detection due to their speed and accuracy Yuan et al. (2022); Zhou et al. (2023); Song et al. (2024). To enhance the performance of the YOLO series in industrial anomaly detection, Yuan et al. (2022) significantly improved the accuracy and efficiency of surface defect detection by optimizing the network structure of YOLOv3-Tiny, improving the loss function, and introducing data augmentation and preprocessing techniques. Zhou et al. (2023) incorporated the CSPlayer module and a global attention mechanism into YOLOv5 to improve the accuracy and speed of small object defect detection on metal surfaces. Song et al. (2024) propose an improved YOLOv8-based model for steel surface defect detection. By integrating deformable convolution, a bidirectional feature pyramid network, and the BiFormer attention mechanism, along with adjusting the loss function to WIoUv3, their model achieves more effective multi-level feature fusion and adaptive attention allocation. These enhancements significantly improve detection accuracy and generalization performance, particularly for defects with complex textures and irregular shapes.

2.1.2 Semi-supervised anomaly detection

Semi-supervised learning is a machine learning approach that integrates collaborative training and prediction using a small amount of labeled data along with a large amount of unlabeled data Cai and Gao (2023). In a semi-supervised framework, a severe imbalance often exists between labeled and unlabeled datasets: labeled defects are typically much fewer than normal samples and fail to cover all defect categories. The mainstream approach first generates confidence-based pseudo labels for unlabeled samples, followed by consistency regularization for knowledge distillation. To prevent further aggravating the imbalance through pseudo labeling, researchers have proposed adaptive re-weighting and uncertainty-aware sampling strategies, which dynamically increase the contribution of high-confidence minority-class samples during training. Additionally, semi-synthetic defect interpolation—such as MixUp or CutMix applied to labeled anomalies—can help augment the apparent variations of rare defects. Semi-supervised learning is particularly suitable for industrial scenarios where labeled data is scarce, defective, or difficult to obtain and annotate. By learning the distribu-

tion of normal samples or leveraging limited labeled abnormal samples, semi-supervised learning can significantly improve the accuracy and adaptability of anomaly detection while reducing the cost of data labeling. For example, Akcay et al. (2019) propose GANomaly, which is based on a conditional generative adversarial network. It achieves efficient anomaly detection by jointly learning image-space generation and latent-space inference. Rudolph et al. (2021) propose a semi-supervised defect detection method called DifferNet to address the challenges posed by the scarcity of defect samples and unknown features in manufacturing. Specifically, DifferNet uses a convolutional neural network to extract image features and a normalizing flow to perform density estimation of multi-scale features. Combined with a multi-projection layer, DifferNet can achieve high-precision detection and localization of defects. Since only a small number of defect samples are required to train the model, DifferNet is of high practical value in large-scale industrial scenarios where defects are difficult to obtain. Yang et al. (2023) propose MemSeg, a semi-supervised learning method for high-precision, real-time surface defect detection of industrial product images. MemSeg uses synthetic and memory samples to assist model learning, and combines multi-scale feature fusion with a spatial attention mechanism. It transforms the semi-supervised detection process into an end-to-end semantic segmentation task, significantly enhancing the model's ability to distinguish between normal and abnormal patterns, thereby achieving efficient and accurate defect detection. Yao et al. (2023) propose BGAD, a boundary-guided adversarial learning method for anomaly detection in semi-supervised settings. This approach enhances the model's discriminative ability by learning the boundaries of known anomalies and effectively mitigates bias introduced by these anomalies.

2.1.3 Self-supervised anomaly detection

Self-supervised learning has emerged as a powerful paradigm for industrial anomaly detection, especially in scenarios where labeled data is scarce, noisy, or expensive to obtain. Unlike traditional supervised methods that rely on detailed annotations, self-supervised approaches generate supervision signals automatically by designing pretext tasks on unlabeled data. Common tasks such as image restoration, contrastive learning, feature enhancement, and sequence prediction enable models to learn robust and informative feature representations, thereby reducing reliance on labeled samples and improving generalization. In industrial contexts, self-supervised learning serves multiple purposes: (1) it enables effective representation learning from abundant normal samples in the absence of labeled anomalies; (2) it facilitates the detection of previously unseen defects by modeling the intrinsic structure of normal data; and (3) it enhances the model's adaptability to diverse and evolving defect patterns in complex real-world environments.

Although current deep learning models perform well on known defect types, their performance often degrades in the presence of unknown defects that did not appear during training. This is mainly because model learning is strongly dependent on the distribution of training data. To address these challenges, researchers have proposed various generalization enhancement strategies. On the one hand, methods based on self-supervised learning, such as CutPaste Li et al. (2021) and PatchSVDD Yi and Yoon (2020), improve the model's ability to identify unknown defects without labels by constructing pseudo-anomaly samples or using local image structure for feature learning. On the other hand, semi-supervised methods such as DifferNet Rudolph et al. (2021) and MemSeg Yang et al. (2023) improve the dis-

criminability of new defects by combining a flow model with a memory mechanism and a small number of known anomalies. Recently, diffusion models with stronger representation capabilities and contrastive learning methods that incorporate linguistic priors (e.g., Win-CLIP Jeong et al. (2023)) have also been widely used to enhance the model's generalization ability to unknown anomalies, providing scalable detection methods for the ever-changing defect morphology in industrial scenarios.

Self-supervised learning is a machine learning method that automatically generates supervision signals by designing pre-training tasks on unlabeled data Hojjati et al. (2024); Li et al. (2024); Luo et al. (2022); Tsai and Wang (2023). It usually uses tasks such as image restoration, contrastive learning, feature enhancement or sequence prediction to learn high-quality feature representations, thereby significantly reducing the dependence on labeled data Wei et al. (2025); Bin and Wang (2024); Huang et al. (2024); Qin et al. (2024); Bauer et al. (2024); Lijie et al. (2024). In industrial anomaly detection, self-supervised learning can achieve efficient anomaly recognition and localization by modeling the features of normal samples, and is especially suitable for application scenarios where abnormal samples are scarce and the data distribution is complex.

During the self-supervised training phase, only normal images are used. However, when constructing pretext tasks, a form of "category" imbalance may arise among different transformation operations (e.g., occlusion, rotation, color perturbation), leading the model to become overly sensitive to certain transformations. To address this, common strategies include introducing memory queue balancing in contrastive learning to ensure that stored feature vectors are evenly distributed across various augmented views, as well as applying random occlusion ratio sampling and transformation-balancing schedules in reconstruction-based tasks. These techniques help the model maintain balanced representation capabilities across diverse potential anomaly patterns. Although such methods do not generate defective samples directly, they can enhance the model's robustness and generalization to various types of anomalies.

Yi et al. (2020) propose PatchSVDD to improve the ability to detect and locate anomalies in images. Unlike traditional SVDD, PatchSVDD uses self-supervised learning to enable the model to automatically aggregate image patches with similar semantics, thereby better-identifying anomalies. In addition, PatchSVDD also introduces a strategy of multi-level coding and multi-scale detection, which further improves the accuracy of anomaly detection by integrating results across different scales. Li et al. Li et al. (2021) propose a self-supervised learning method called CutPaste, which aims to train high-performance defect detection models using only normal sample data. The method constructs "pseudo-anomaly" samples for training by cutting out parts of normal images and randomly pasting them onto another image to simulate defects. CutPaste adopts a two-stage framework: first, a deep representation of normal data is learned, and then a generative one-class classifier is used to distinguish normal data from data containing abnormal patterns. In addition, CutPaste can extend the learning to the image patch level, so that the defective areas in high-resolution images can be accurately located without using labeled data. Zou et al. (2022) propose a self-supervised pre-training method SPot-the-Difference (SPD) to address the problem of insufficient local sensitivity in existing self-supervised learning for anomaly detection and segmentation tasks. By introducing the SmoothBlend local enhancement technique and the SPD training strategy, the method effectively improves the model's local sensitivity to abnormal patterns, significantly enhancing the performance of anomaly detection and

segmentation. Ristea et al. (2022) propose a new self-supervised pre-trained convolutional attention module (SSPCAB) for effectively reconstructing and identifying anomalous samples. This method trains the model by masking or removing portions of input data and predicting the missing information, using reconstruction error as an anomaly metric. Madan et al. (2023) propose the self-supervised masking convolutional transformer block (SSMCTB) to address the shortcomings of traditional anomaly detection methods in capturing global structural features. SSMCTB achieves anomaly detection by masking input information at any layer of the neural network and reconstructing the masked data using 3D convolutions and transformer blocks, thus only learning from normal samples and not relying on anomalous samples. Cai et al. (2023) present an innovative and robust anomaly detection framework, DAF, which improves on the existing self-supervised learning methods that rely too heavily on the appearance of synthetic data. Although traditional self-supervised methods perform well on anomaly detection tasks, their sensitivity to synthetic data often limits their generalization ability in practical applications. To this end, DAF introduces a teacher-student network to generate appearance-independent difference maps as guidance, and combines a segmentation decoder with an auxiliary head to accurately localize anomalies.

Tien et al. (2023) review the reverse distillation Deng and Li (2022) technique and propose the RD++ method, which simulates pseudo-anomalous images by adding random noise to normal images to enhance the model's generalization ability to anomalies. RD++ introduces a multi-projection layer on this basis to learn feature representations from different scales. Additionally, a self-supervised optimal transport loss is used to further compress the feature space, making the feature distribution of normal images more compact. Finally, RD++ combines contrastive loss and reconstruction loss under a multi-task learning framework, which not only effectively compresses the feature distribution of normal samples, but also suppresses abnormal signals, thereby achieving high-precision and real-time inference capabilities for anomaly detection and localization on multiple datasets. Yoo et al. (2023) propose ST-SSAD, a self-supervised outlier detection framework for systematically tuning the hyperparameters of data augmentation in an unsupervised setting. The method combines data augmentation strategies with the model tuning process and enables automatic and effective parameter tuning of outlier detection models via an unsupervised validation loss and a differentiable augmentation function. As it does not rely on label information, ST-SSAD is particularly suitable for outlier detection tasks where labels are difficult to obtain. Zuo et al. (2024) propose ReconFA, a self-supervised domain adaptation method that aims to solve the problem of mismatched pre-trained CNN features and the object data domain. By designing a self-supervised multi-scale feature fusion strategy that combines a multi-scale aggregation module with a feature reconstruction task, ReconFA can effectively adapt pre-trained features to the object domain, thereby significantly improving the performance of anomaly detection. Li et al. (2024) propose IMRNet, a self-supervised learning method that aims to solve the current problem of limited model scalability in 3D anomaly detection due to insufficient real data. The method effectively detects and localizes abnormal regions in 3D point clouds through iterative mask reconstruction and feature comparison. Zhang et al. (2024) propose RealNet, a feature selection network for generating realistic and diverse synthetic anomaly samples in industrial image anomaly detection to address the difficulties faced by existing detection methods. RealNet consists of three modules: strength-controllable diffusion anomaly synthesis (SDAS) is used to generate synthetic samples with different anomaly intensities; anomaly-aware features selection (AFS) helps select a representative

subset of pre-trained features; and reconstruction residuals selection (RRS) adaptively screens reconstruction residuals that can assist in locating anomalous regions. Through the cooperation of these three modules, RealNet achieves outstanding performance in anomaly detection tasks. Tu et al. (2024) propose local-to-global self-supervised feature adaptation (LSFA) to fine-tune adapters and learn task-oriented representations for anomaly detection through local-to-global self-supervised feature adaptation. The core of LSFA lies in solving the problem of domain gaps between pre-trained models and object industrial and source data, optimizing the feature representation through in-module feature compactness (IFC) optimization and cross-modal local-to-global consistency (CLC) alignment. IFC optimizes the compactness of features at the local (patch-wise) and global (prototype-wise) levels through a dynamically updated memory bank, making normal features more concentrated thus improving the sensitivity to abnormal features. CLC, on the other hand, aligns features from different modalities (RGB images and 3D point clouds) through contrastive learning, enhancing the interaction of multimodal information and ensuring the consistency of features from different modalities at the local and global levels. LSFA uses pre-trained vision and point transformer as feature extractors, and introduces an adapter to fine-tune the pre-trained features to adapt to specific anomaly detection tasks.

Overall, self-supervised learning offers an effective solution for reducing annotation costs while enabling scalable and generalizable anomaly detection by leveraging the intrinsic structure of normal data. These advantages make it particularly valuable in industrial scenarios characterized by high variability and limited supervision.

2.1.4 Weakly supervised anomaly detection

Weakly supervised learning can effectively reduce labeling costs while reducing the dependence on finely labeled data, and improve the automation and accuracy of detection. It is particularly advantageous in scenarios with complex data distributions and scarce abnormal samples, offering an efficient and flexible solution for real-world industrial applications Shenxing et al. (2023); Wang et al. (2024). Weakly supervised methods rely solely on image-level OK/NG labels, resulting in extremely imbalanced positive and negative samples and a lack of precise localization information. To reduce the model's bias toward normal samples, one common approach is multi-instance learning with heatmap reweighting: higher weights are assigned to the highest-scoring patches in NG images, encouraging the model to focus on potential defect regions. Alternatively, pseudo-defects can be synthesized at the local level to enrich the diversity of negative samples. In parallel, pseudo-label re-estimation can iteratively refine the defect regions within NG samples, progressively balancing the number of effective positive and negative patches and enhancing localization accuracy. Ding et al. (2022) propose DRA, an anomaly detection method based on decoupled representation learning, designed to detect both known and unknown anomaly types with only a limited number of labeled anomaly samples. DRA significantly improves the model's generalization ability for various anomalies by learning multiple decoupled anomaly representations, including known anomalies, pseudo-anomalies, and potential residual anomalies. This leads to effective detection and overall excellent performance when facing different types of anomalies. Zhang et al. (2023) propose the prototypical residual network (PRN) to address the problems of data scarcity, subtle and difficult-to-detect anomalies, and appearance variability in industrial anomaly detection and localization. Specifically, PRN learns the feature

residuals between abnormal and normal patterns, and reconstructs the segmentation map of abnormal regions by combining multi-scale prototypes and multi-dimensional self-attention mechanisms, so as to achieve accurate detection and localization of abnormal regions of different sizes, shapes and quantities. In addition, PRN also incorporates diverse anomaly generation strategies to expand and diversify abnormal samples, further enhancing model generalization and ultimately achieving efficient detection and accurate localization. Xie et al. (2024) propose the IM-IAD benchmark to address the lack of a unified evaluation benchmark in the field of industrial image anomaly detection. IM-IAD evaluates the performance of various algorithms under multiple levels of supervision, different learning paradigms, and efficiency requirements, and conducts large-scale experiments on seven datasets and multiple algorithms, providing a valuable reference for improving industrial image anomaly detection algorithms and identifying follow-up research directions.

Zhu et al. (2024) propose abnormal heterogeneity learning (AHL), which aims to detect and identify previously unseen anomalies during training, even with only a few anomaly samples in the open set anomaly detection scenario. AHL addresses the generalization limitations of existing methods, which struggle due to their assumption that abnormal samples follow a homogeneous distribution. By simulating a diverse abnormal distribution and collaboratively training a differentiable model, AHL enables the learning of a unified model applicable to various new domains while efficiently detecting different types of heterogeneous anomalies. Cao et al. (2024) propose BiasS, a framework for anomaly detection that aims to address the problem that unsupervised image anomaly localization methods are prone to overfitting in supervised tasks and cannot effectively utilize seen anomaly information. BiasS first generates bias knowledge about known and unknown anomalies, which is then passed to two specialized student networks for detection. By elevating unsupervised methods to a level similar to supervised learning, BiasS improves the performance and generalization ability of unsupervised methods in open world anomaly detection. Rolih et al. (2024) propose the tiling ensemble (TE) method, which specifically addresses the high memory usage problem faced by high-resolution industrial anomaly detection when detecting small-sized anomalies. TE divides high-resolution images into multiple smaller tiles and trains an independent model for each tile, significantly reducing memory consumption. Additionally, TE also introduces an overlapping division strategy to improve anomaly detection performance. Bovzirc et al. (2021) propose a hybrid supervision method that combines fully supervised and weakly supervised approaches to reduce reliance on fine annotations and flexibly adapt to varying levels of annotation granularity. The method employs two sub-networks: one for pixel-level defect segmentation and the other for image-level defect classification. It achieves detection performance comparable to that of fully supervised models, even when relying only on weak annotations, while significantly lowering the dependence on accurate annotations and reducing annotation costs.

2.1.5 Unsupervised anomaly detection

Compared with supervised learning, which requires annotated abnormal samples, unsupervised anomaly detection identifies anomalies by uncovering patterns within the data itself, without relying on any defect labels. The training data may consist of purely normal samples or mixed samples without distinction between normal and abnormal. A wide range of methods are employed, including density estimation, clustering, reconstruction-based gen-

eration, and contrastive learning. With strong generalization ability, unsupervised methods can detect previously unseen defects. Given these unique features, unsupervised learning has found valuable applications in industrial visual inspection. In this context, unsupervised learning identifies anomalies using unlabeled data, extracting distribution patterns and feature representations from the dataset Qiangwei et al. (2024); Haohao et al. (2023). A common approach involves constructing a normal pattern representation based on statistical properties of normal samples or generative models and detecting anomalies as deviations from this pattern. Unsupervised learning is particularly suitable for detection tasks with a large number of normal samples and relatively few abnormal samples, and has demonstrated a high degree of automation and excellent detection results in practical applications. These characteristics and advantages make unsupervised learning an indispensable tool in the field of industrial anomaly detection. It is also important to note that dataset imbalance—namely the overwhelming predominance of normal samples over anomalies—has a significant impact on model training and performance. In industrial visual anomaly detection, this imbalance can lead to biased decision boundaries, degraded detection sensitivity for rare events, and high false negative rates. Unsupervised methods are often considered suitable for such settings, as they typically assume that only normal data is available during training. However, the absence of anomalous samples can limit the model's ability to generalize to subtle or unknown defects. To alleviate this, many recent approaches incorporate pseudo-anomalies, anomaly synthesis, or contrastive learning strategies to balance the latent feature space. These techniques aim to mimic the characteristics of rare anomalies, improve the separability of learned representations, and ultimately enhance detection robustness in highly imbalanced environments. In industrial visual anomaly detection, data augmentation plays a crucial role, especially under the constraint of limited anomalous samples. Effective augmentation strategies can significantly enhance the model's generalization and robustness by simulating realistic anomaly scenarios. For example, Patch-wise Augmentation Fan et al. (2024) reformulates unsupervised anomaly detection as a supervised object detection task by synthesizing similar anomalies through self-supervised learning, thus enabling direct localization of anomalies. RealNet Zhang et al. (2024) introduces a strength-controllable diffusion anomaly synthesis (SDAS) module to generate diverse and realistic synthetic anomalies, enriching the training distribution. Compared to traditional augmentation techniques (e.g., flipping, cropping), these anomaly-aware strategies focus more on the local variability and contextual consistency of defective regions.

Wang et al. (2021) propose using both local and global feature comparisons to address the problem of obtaining and labeling a large amount of abnormal data in industrial scenarios. Specifically, the method uses Local-Net to extract the local features and Global-Net to obtain the global features. In addition, two detection heads are introduced: inconsistency anomaly detection, which is used to capture inconsistencies between global and local features, and distortion anomaly detection, which is used to detect subtle defects within patches. By combining feature information at both local and global levels, the method effectively detects various types of anomalies without requiring many annotated anomaly samples.. Yu et al. (2021) propose FastFlow, which utilizes a lightweight 2D normalized flow model to map the features of the input image to a standard normal distribution. The method then uses the probability value as the anomaly score to efficiently and accurately detect and localize anomalies. McIntosh et al. McIntosh and Albu (2023) propose InReaCh, which is designed to eliminate the reliance of traditional image anomaly detection meth-

ods on purely normal training data. Specifically, InReaCh constructs multiple channels by associating high-confidence normal image patches in the training set and then uses these channels to evaluate image patches in new inputs. This enables efficient anomaly detection and localization in complex images containing anomalies. Liang et al. (2023) propose OCR-GAN to address the performance bottleneck caused by the insufficient reconstruction ability of traditional reconstruction methods. Specifically, OCR-GAN introduces an innovative Frequency Decoupling (FD) module to decompose the input image into different frequency distributions. Additionally, a Channel Selection (CS) module facilitates information exchange among multiple frequency branches, significantly enhancing the performance of reconstruction-based unsupervised anomaly detection. Zhou et al. (2024) propose MSFlow, a multi-scale flow model framework designed to address the challenges of unsupervised anomaly detection caused by variations in object size. MSFlow employs heterogeneous parallel flows and fusion flows to capture features and exchange information across different scales, thereby significantly enhancing performance and generalization in both image-level and pixel-level anomaly detection and localization tasks. Yang et al. (2024) propose reconstruction as sequence (RAS), a method designed to address the issue of insufficient context awareness in existing feature reconstruction approaches for unified unsupervised anomaly detection. RAS is built upon the transformer framework and introduces the RAS-Transformer module to enhance context correspondence during reconstruction. Chen et al. (2024) propose PBAS, a novel progressive boundary-guided anomaly synthesis strategy that synthesizes key feature-level anomalies in a directed manner without relying on auxiliary textures. PBAS comprises three core components: approximate boundary learning (ABL), anomaly feature synthesis (AFS), and refined boundary optimization (RBO). To make the distribution of normal samples more compact, ABL first learns an approximate decision boundary through a center constraint, and improves center initialization via feature alignment. Then AFS synthesizes anomalies with more flexible scales guided by the hypersphere distribution of normal features. Since the initial boundary is loose and may contain real anomalies, RBO refines the decision boundary by performing binary classification between artificial anomalies and normal features. Radicioni et al. (2025) accurately estimated changes in chain pitch through power spectral density analysis of vibration signals. They combined these findings with local temperature measurements to evaluate sliding rail wear. This approach provides robust technical support for improving the reliability and efficiency of industrial production.

As shown in Fig. 2, this subsection provides a detailed overview of industrial visual anomaly detection methods. Section 2.2 describes feature embedding-based approaches, including teacher-student, one-class classification, distribution-map, memory bank, contrastive language-image pre-training and anomaly clustering. Then, Sect. 2.3 introduces the reconstruction-based methods, including autoencoder, generative adversarial networks, transformer, diffusion model, generative pre-training transformer, segment anything model. Finally, Sect. 2.4 presents 3D anomaly detection.

2.2 Feature embedding-based methods

In industrial visual anomaly detection, feature embedding-based methods effectively distinguish abnormal samples by mapping image data into a feature space. This section provides a comprehensive overview of feature embedding-based methods for industrial visual anomaly

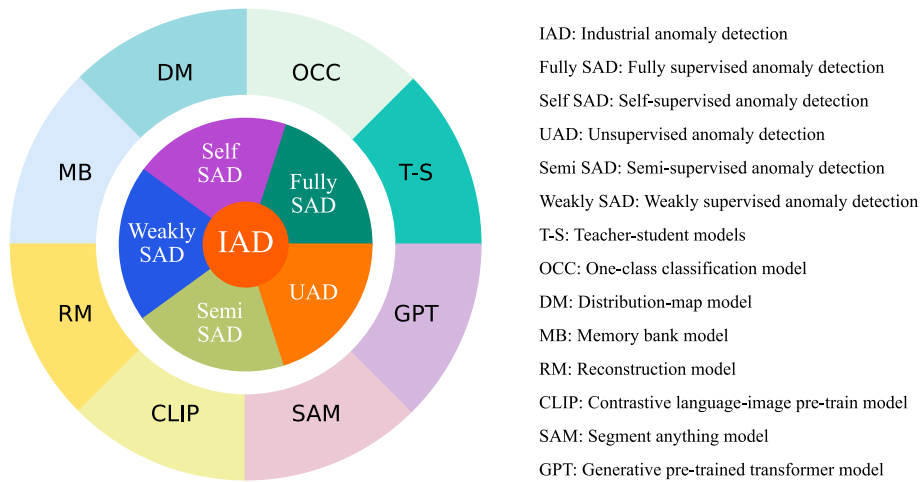


Fig. 2 Overview of the mapping between learning strategies and representative method families in industrial visual anomaly detection

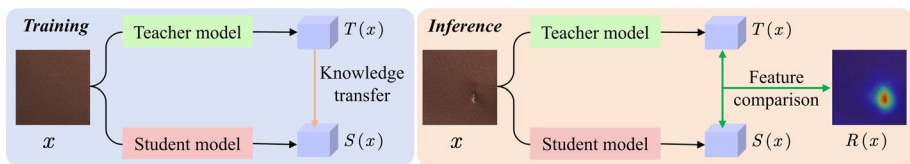


Fig. 3 Teacher-student model. **Training phase:** The student model iteratively trains to learn and approximate the teacher model's feature representations, enabling knowledge transfer. **Inference phase:** The teacher model serves as the baseline, and its features are compared with those of the student model. When the input contains anomalies, the discrepancy between the teacher and student features becomes significant, thereby facilitating the localization and detection of anomalous regions

detection. We thoroughly describe teacher-student models (Sect. 2.2.1), one-class classification (Sect. 2.2.2), distribution maps (Sect. 2.2.3), memory banks (Sect. 2.2.4), contrastive language-image pre-training 2.2.5 and anomaly clustering 2.2.6.

2.2.1 Teacher-student

As shown in Fig. 3, the teacher-student network-based industrial anomaly detection method integrates transfer learning and knowledge distillation. The teacher network is typically a large, fully trained model that has learned a rich representation of normal data features from a large dataset. The student network is smaller in size and is trained by imitating the output of the teacher network. During the training phase, the teacher network is first trained using a dataset containing only normal samples. Subsequently, the student network learns the feature distribution of the teacher network through a knowledge distillation process, with the intermediate output or feature map of the teacher network as the learning object. During the inference phase, the industrial image to be detected is input into both the teacher and student network, and the difference between the two outputs is compared. If the difference is signifi-

cant, it indicates that an abnormality may exist in the image. The student network is small, which improves computational efficiency and reduces resource consumption. Meanwhile, since the teacher network has learned a rich feature representation, the student network can better capture and identify anomalies, enhancing detection accuracy.

The teacher model T is a pre-trained, high-performance network, while the student model S is a smaller network designed to learn and approximate the feature output of the teacher model. Given an input industrial image I , the teacher and student model extract its features, denoted as F^T and F^S , respectively.

$$\begin{cases} F^T = T(I; \theta_T), \\ F^S = S(I; \theta_S). \end{cases} \quad (1)$$

where $F^T \in \mathbb{R}^{H \times W \times C_T}$, $F^S \in \mathbb{R}^{H \times W \times C_S}$ are the output feature maps, with height H and width W . C_T and C_S represent the number of feature channels. The parameter θ_T of the teacher model is fixed and not updated during training, while θ_S represents the trainable parameters of the student model, which are optimized during training. The goal of the training phase is to minimize the difference between F^T and F^S . Common metrics used to quantify this difference include mean squared error (MSE) and cosine similarity loss:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{H \times W \times C_T} \sum_{i=1}^H \sum_{j=1}^W \sum_{k=1}^{C_T} (F_{i,j,k}^T - F_{i,j,k}^S)^2, \quad (2)$$

$$\mathcal{L}_{\text{cos}} = 1 - \frac{\sum_{i,j,k} F_{i,j,k}^T \cdot F_{i,j,k}^S}{\|F^T\| \cdot \|F^S\|}. \quad (3)$$

As shown in Eq. 4, sometimes it is possible to combine multiple losses into a composite loss:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{MSE}} + \beta \mathcal{L}_{\text{cos}}, \quad (4)$$

where, α and β are the weighting parameters. In the inference stage, an anomaly score is computed by measuring the difference between the teacher and student model outputs. The feature difference at each position (i, j) , denoted as $D_{i,j}$ is defined as:

$$D_{i,j} = \|F_{i,j}^T - F_{i,j}^S\|_2, \quad (5)$$

where $F_{i,j}^T$ and $F_{i,j}^S$ are the feature vectors of the feature map at position (i, j) . Map all positional differences $D_{i,j}$ to an anomaly distribution map $D = \{D_{i,j} | i = 1, 2, \dots, H, j = 1, 2, \dots, W\}$, where $D \in \mathbb{R}^{H \times W}$. Globally normalize the anomaly distribution map D to compute the overall anomaly score:

$$\mathcal{A} = \frac{1}{H \times W} \sum_{i,j}^H D_{i,j}, \quad (6)$$

The image is classified as abnormal if the anomaly score $\mathcal{A} > \tau$, where τ is the anomaly detection threshold.

Specifically, Bergmann et al. (2020) propose uninformed students (US), an unsupervised anomaly detection method based on the teacher-student framework. US trains a student network to mimic the output of a pre-trained teacher network and detects anomalies by analyzing the prediction uncertainty and regression error of the student network. Wang et al. (2021) propose an unsupervised anomaly detection method that simultaneously compares local and global features. The method uses a local-net to extract features from image patches and a global-net to capture contextual features from the surrounding environment. Additionally, it introduces inconsistency anomaly detection and distortion anomaly detection heads to detect mismatches between local and global features and subtle defects within patches, respectively. Salehi et al. (2021) propose multiresolution knowledge distillation (MKD) to learn representations that distinguish normal and anomalous samples when only normal samples are available. MKD leverages a pre-trained expert network on ImageNet and distills its features from multiple intermediate layers into a simpler clone network. Anomalies are detected and localized by analyzing the feature differences between the expert and clone networks when processing anomalous inputs. Deng et al. (2022) propose an anomaly detection method called reverse distillation (RD). RD employs teacher and a student encoders, where the teacher extracts a single-class embedding, and the student reconstructs a multi-scale representation. To improve anomaly detection accuracy, RD introduces a trainable one-class bottleneck embedding module, which compresses high-dimensional features into compact embeddings, preserving normal pattern information while suppressing anomalous disturbances. RD performs knowledge distillation on multi-scale features, utilizing cosine similarity loss to quantify activation differences between the teacher and student models. The loss function is computed by aggregating multi-scale anomaly maps, enabling the student network to mimic the teacher network's behavior and achieve high-performance anomaly detection. Wan et al. (2022) propose an unsupervised image anomaly detection and segmentation method based on pre-trained feature mapping (PFM). PFM detects anomalies by mapping images from one pre-trained feature space to another. To further enhance performance, PFM introduces bidirectional PFM (B-PFM) and multi-level bidirectional PFM (MB-PFM). B-PFM incorporates a reverse mapping mechanism to enable bidirectional feature utilization, significantly improving information efficiency and model performance. MB-PFM integrates multi-level feature mapping, enhancing the model's ability to detect complex anomalies through multi-scale feature fusion. Yamada et al. (2022) propose student-teacher pyramid matching (STPM), which enhances accuracy by integrating a student-teacher network with a discriminative network. Specifically, the student network reconstructs normal image features, while the teacher network provides feature representations for anomalous regions. Additionally, STPM employs a discriminative network trained on pseudo-anomalies generated from anomalous images, enabling more precise differentiation between normal and anomalous regions. Rudolph et al. Rudolph et al. (2023) propose the asymmetric student-teacher network (AST), which employs normalized flows as the teacher network for density estimation and trains a feed-forward network as the student network to detect anomalies at larger distances. Cao et al. (2022) propose an anomaly segmentation method based on information knowledge distillation (IKD). IKD addresses overfitting issues arising when a neural network's capacity does not align with its knowledge by extracting informative knowledge and providing strong supervisory signals. To enhance

segmentation performance, IKD incorporates context similarity loss (CSL) and adaptive hard sample mining (AHSM). CSL captures contextual information within the normal data manifold. AHSM prioritizes hard samples that contain valuable information, encouraging the model to focus on more challenging cases. Gu et al. (2023) propose memory-guided knowledge distillation (MemKD), which enhances the normality of features generated by the student model by storing normal data information. Additionally, MemKD employs a normalcy representation learning strategy to construct a normal sample set, enabling the normal knowledge recall module to learn prior knowledge without anomalous data and retrieve it during inference. Zhang et al. (2023) propose DeSTSeg to address the lack of constraints on anomalous data and the empirical nature of multi-level feature fusion in traditional teacher-student anomaly detection frameworks. DeSTSeg integrates a pre-trained teacher network, a denoising student encoder-decoder, and a segmentation network. By introducing a denoising process, the student network generates feature representations that differ from those of the teacher network for anomalous inputs. Additionally, the segmentation network adaptively fuses multi-level features, significantly enhancing anomaly detection performance. Tong et al. (2024) propose EMMFRKD, an enhanced anomaly detection and localization model based on inverse distillation. EMMFRKD extracts discriminative features across multiple scales by incorporating an enhanced multi-scale feature mutual mapping fusion module. Additionally, it employs a coordinate attention mechanism and a single-category embedding memory bank to enhance the representation of abnormal regions and improve model reconstruction quality. Wu et al. (2024) propose autoencoder knowledge distillation (AEKD), which addresses the model generalization problem in traditional knowledge distillation methods by introducing an asymmetric teacher-student model structure. Additionally, it incorporates a trainable multi-scale feature fusion module to mitigate anomaly interference and designs distinct data flows for the teacher and student models, enhancing their differences in anomaly representation. Gu et al. (2024) propose ASTMFR, a novel asymmetric student-teacher network framework designed to address the challenge of collecting anomalous samples. ASTMFR enhances the feature alignment of the student network on normal samples by introducing a masked feature regeneration (MFR) module. Additionally, it leverages an asymmetric network structure to increase the feature diversity of unseen anomalous samples during testing. This approach mitigates the issue of high feature similarity among anomalous samples in traditional methods, significantly improving anomaly detection and localization accuracy.

Li et al. (2024) propose MCAD, a multi-class anomaly detection framework. Specifically, MCAD integrates relational knowledge distillation in the distillation process to enhance knowledge transfer and improve the expressiveness of student networks, so as to achieve high-performance anomaly detection, localization, and classification. Zhang et al. (2024) propose DSKD, a dual-student knowledge distillation framework for structural and logical anomaly detection. Specifically, the local student focuses on accurate low-level feature reconstruction, primarily for detecting structural anomalies, while the global student learns global correlations to enhance logical anomaly detection. To improve global information capture, the global context concentration block (GCCB) is introduced in the global student. GCCB retains the most critical global information by compressing high-level semantic features and restoring them to their original size. To further enhance the global student's ability to learn global context, a context affinity loss is introduced. This loss is computed by measuring the cosine similarity between the student's feature vector and all feature vectors

in the teacher feature map, converting the results into a probability distribution. The KL divergence quantifies the difference between the teacher and student distributions, optimizing the global student's feature learning. Batzner et al. (2024) propose EfficientAD, a visual anomaly detection method designed to balance computational efficiency and detection performance in existing anomaly detection approaches. EfficientAD introduces a lightweight feature extractor and a student-teacher network to detect abnormal features and identify logical anomalies. The student-teacher network, inherited from the encoder, enables high-accuracy anomaly detection while maintaining low latency and high throughput. EfficientAD achieves precise detection of structural and logical anomalies with millisecond latency. Zhang et al. (2024) propose FCACDL, a feature distillation learning method designed to address the limitations of teacher-student networks in feature representation and anomaly detection performance constrained by the teacher network. FCACDL employs a vision transformer as the backbone network and integrates the central feature strategy, self-attention mask strategy, and multi-view consistency constraint. By leveraging feature distillation and self-supervised learning, it enhances visual anomaly detection performance. Liu et al. (2024) propose dual-modeling decoupling distillation (DMDD) to address the overgeneralization problem in existing knowledge distillation methods for anomaly detection. DMDD enhances anomaly detection accuracy and robustness by constructing a decoupled student-teacher network and introducing a dual-mode distillation strategy, consisting of normal distillation and anomalous distillation. Additionally, it employs a multi-sensory segmentation network to optimize anomaly graph fusion.

2.2.2 One-class classification

One-class classification serves as a transitional paradigm between self-/unsupervised and weakly supervised learning. During the training phase, only normal samples are involved—similar to unsupervised learning—but the objective function explicitly learns a single-class boundary to distinguish potential anomalies, rather than merely performing reconstruction or density estimation. This approach offers labeling costs comparable to those of unsupervised learning, while delivering discriminative performance close to that of supervised methods, making one-class classification one of the most cost-effective strategies in industrial applications. As shown in Fig. 4, one-class classification refers to a classification problem where only one class is known during training. It learns the feature space distribution of data points in this class to establish a model for determining whether new data points belong to the same class. One-class classification focuses solely on identifying data points within a single class, without requiring prior knowledge of multiple classes. The core idea of OCC

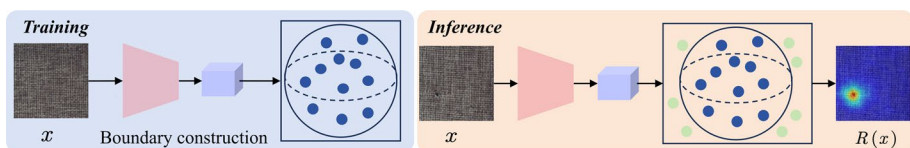


Fig. 4 One-class classification method. **Training phase:** The model is trained exclusively on normal samples, constructing a boundary within the feature space that encloses these normal data points. **Inference phase:** The validation data undergo the same feature extraction process to produce feature vectors, which are then compared against the previously learned boundary. Any point falling within the boundary is deemed normal; otherwise, it is classified as anomalous

is to learn a boundary or probability distribution that encapsulates the feature space of normal samples, identifying samples that deviate from this boundary as anomalous. Common OCC methods include boundary-based and probability distribution-based models. Among these, Support Vector Data Description (SVDD) is a representative algorithm that classifies new data points by constructing a minimal hypersphere enclosing all training samples: data points inside the hypersphere are considered normal, while those outside are deemed anomalous. For an input industrial image I , the feature representation f is defined as:

$$f = E(I; \theta_E), \quad (7)$$

where E is the feature extraction network, and θ_E represents its parameters. $f \in \mathbb{R}^d$ denotes the feature vector of the image, where d is the feature dimension. The goal of SVDD is to find a hypersphere with minimal radius R that encloses the feature vectors of normal samples.

$$\min_{R, c} R^2 + \frac{1}{\nu N} \sum_{i=1}^N \xi_i, \quad (8)$$

where $\|f_i - c\|^2 \leq R^2 + \xi_i$, $\xi_i \geq 0$, $\forall i$, R and c are the radius and center of the hypersphere, respectively. ξ_i is the slack variable, allowing some samples to lie outside the hypersphere. ν is the tuning parameter that controls the weight of the slack variable. N represents the number of normal samples. The anomaly score is defined as the distance between a sample and the center:

$$\mathcal{A}_{distance}(f) = \|f - c\|^2, \quad (9)$$

where $\mathcal{A} > R^2$, the sample is abnormal.

Similarly, in autoencoder-based OCC, the model learns to reconstruct normal samples through an encoder-decoder structure. The objective of the autoencoder is to minimize the reconstruction error:

$$\mathcal{L}_{recon} = \frac{1}{N} \sum_{i=1}^N \|f_i - \hat{f}_i\|^2, \quad (10)$$

where f_i is input feature, and $\hat{f}_i$ is the reconstructed feature. During the inference stage, the reconstruction error is computed as the anomaly score:

$$\mathcal{A}_{recon}(f) = \|f - \hat{f}\|^2, \quad (11)$$

A sample is classified as anomalous if $\mathcal{A}(f) > \epsilon$, where ϵ is the threshold for the reconstruction error. A comprehensive anomaly score can be obtained by combining multiple scoring methods, such as boundary distance and reconstruction error:

$$\mathcal{A}_{final} = \alpha \mathcal{A}_{distance} + \beta \mathcal{A}_{recon}, \quad (12)$$

where, $\mathcal{A}_{\text{distance}} = \|f - c\|^2$ represents the SVDD-based anomaly score. $\mathcal{A}_{\text{recon}} = \|f - \hat{f}\|^2$ represents the autoencoder-based anomaly score. α and β are weighting parameters.

Specifically, Li et al. (2021) propose CutPaste, a self-supervised learning method designed to train high-performance defect detection models using only normal images. The method simulates anomalies by cutting patches from a normal image and pasting them at random locations within the same image, thereby introducing local structural inconsistencies. Cut-Paste follows a two-stage framework: in the first stage, it learns deep feature representations of normal data; in the second stage, it constructs a one-class classifier to distinguish between normal patterns and those altered by synthetic anomalies. Furthermore, the method extends to patch-level representation learning, which enables effective defect localization in high-resolution images without requiring any labeled data during training. Cao et al. (2023) present ADShift, a generalized normal learning method designed to address anomaly detection under distribution shifts. ADShift enhances the accuracy and robustness of anomaly detection by minimizing the distribution gap between in-distribution and out-of-distribution normal samples during both training and inference in an unsupervised manner. This approach effectively mitigates the impact of distribution inconsistency. Liu et al. (2023) propose SimpleNet, which consists of four key components: a pre-trained feature extractor, a feature adapter, an anomaly feature generator, and an anomaly discriminator. SimpleNet generates anomalous features by introducing Gaussian noise into the feature space and employs a binary anomaly discriminator to distinguish between normal and anomalous features, thereby enabling efficient and accurate anomaly detection and localization. Yang et al. (2024) propose SLSG, a semi-supervised anomaly detection method for industrial images, designed to address challenges in feature embedding and one-class classification. SLSG integrates self-supervised learning and self-attention convolutional networks, leveraging a generative pre-trained network to assist the encoder in learning feature embeddings of normal patterns. Additionally, it introduces simulated anomalous samples to provide pseudo-prior knowledge of anomalies and constructs a general graph structure to model relationships between image elements comprehensively. SLSG significantly enhances industrial image anomaly detection performance, particularly in logical anomaly detection. Strater et al. (2024) propose GeneralAD, an anomaly detection framework designed to overcome the limitations of existing methods in cross-domain applications. GeneralAD leverages a visual transformer for feature extraction and integrates a self-supervised anomaly feature generation module to create hard-to-recognize pseudo-anomalies. It then employs an attention-based discriminator to accurately detect image-level anomalies and generate interpretable anomaly maps. GeneralAD enables generalized anomaly detection across semantic, near-distribution, and industrial settings. He et al. (2024) propose DS2, a two-stage intensive pre-training model designed to address the poor transferability of models in image anomaly localization caused by the lack of comprehensive anomaly samples. DS2 integrates double-positive sample selection criteria and multi-scale feature learning to enhance the model's ability to distinguish small and subtle anomalies effectively. Compared to end-to-end models, DS2 demonstrates greater transferability on new datasets and exhibits higher stability and robustness, even with small training datasets and small batch sizes. Chen et al. (2024) propose GLASS, a unified anomaly detection framework designed to overcome the limitations of existing strategies in industrial anomaly detection, particularly in anomaly coverage and controllability. It specifically addresses the challenge of detecting weak defects that closely resemble normal regions. GLASS improves weak defect detection by integrating global anomaly syn-

thesis (GAS) and local anomaly synthesis (LAS). Additionally, it employs gradient ascent and truncated projection-guided Gaussian noise to generate near-distribution anomalies in a controllable manner, further enhancing defect detection. Furthermore, GLASS synthesizes diverse anomalies at the image level through texture superposition, achieving a broader and more controllable anomaly synthesis strategy, which significantly enhances industrial anomaly detection and localization performance.

2.2.3 Distribution-maps

As shown in Fig. 5, the distribution-map method constructs a distribution map of normal samples in the feature space. It measures how much a sample's features deviate from the normal distribution and generates an anomaly distribution map to identify abnormal regions.

A pre-trained feature extraction network is used to transform an industrial image I into a feature map F . Let the dimension of the feature map be $F \in \mathbb{R}^{H \times W \times C}$, where H and W are the height and width of the feature map, respectively. C is the number of channels. For each feature vector $f_{i,j} \in \mathbb{R}^C$ at position (i, j) , define:

$$F = \{f_{i,j} | i = 1, 2, \dots, H, j = 1, 2, \dots, W\}, \quad (13)$$

The characteristic distribution of a normal sample is typically modeled using a Gaussian distribution to describe the spatial distribution of features. Assume that the feature vector $f_{i,j}$ of a normal sample follows a Gaussian distribution at each position (i, j) .

$$f_{i,j} \sim \mathcal{N}(\mu_{i,j}, \sum_{i,j}), \quad (14)$$

where $\mu_{i,j} \in \mathbb{R}^C$ is the mean vector at position (i, j) , and $\sum_{i,j} \in \mathbb{R}^{C \times C}$ is the covariance matrix. During the training phase, the mean vector $\mu_{i,j}$ and the covariance matrix $\sum_{i,j}$ are estimated using a normal sample:

$$\mu_{i,j} = \frac{1}{N} \sum_{n=1}^N f_{i,j}^n, \quad \Sigma_{i,j} = \frac{1}{N} \sum_{n=1}^N (f_{i,j}^n - \mu_{i,j})(f_{i,j}^n - \mu_{i,j})^T, \quad (15)$$

where N denotes the number of normal samples. $f_{i,j}^n$ represents the feature vector of the n -th normal sample at position (i, j) . For the feature vector $f_{i,j}$ of the input sample, compute its

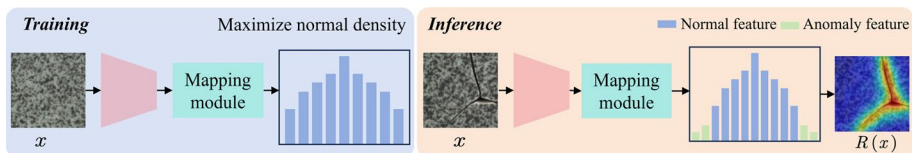


Fig. 5 Distribution-map method. **Training phase:** The model extracts features from normal samples and, via a mapping module, projects the input data into a feature distribution space with the aim of maximizing the density of normal data. **Inference phase:** The same mapping module is used to map incoming samples into the same feature space. The sample is classified as anomalous when their feature distribution differs substantially from the previously learned normal distribution

Mahalanobis distance from the normal distribution to quantify the degree of abnormality at that location:

$$D_{i,j} = \sqrt{(\mathbf{f}_{i,j} - \mu_{i,j})^T \sum_{i,j}^{-1} (\mathbf{f}_{i,j} - \mu_{i,j})}. \quad (16)$$

A larger $D_{i,j}$ indicates a greater deviation from the normal distribution, suggesting a higher likelihood of abnormality at that position. Map the Mahalanobis distance $D_{i,j}$ at all positions to a distribution map $\mathbb{D} \in \mathbb{R}^{H \times W}$:

$$\mathbb{D} = \{D_{i,j} | i = 1, 2, \dots, H, j = 1, 2, \dots, W\}. \quad (17)$$

By setting a threshold ϵ , the abnormal regions in the distribution map can be identified.

$$A_{i,j} = \begin{cases} 1, & \text{if } D_{i,j} > \epsilon, \\ 0, & \text{otherwise.} \end{cases} \quad (18)$$

where $A_{i,j}$ is a binary anomaly mask, where 1 indicates an anomalous region and 0 represents a normal region. ϵ is the threshold, determined based on the training set or the validation set. To compute the anomaly score $\mathcal{A}$ for the entire input image, the distribution map $\mathbb{D}$ is normalized and summed:

$$\mathcal{A} = \frac{1}{H \times W}. \quad (19)$$

Set the global threshold τ . If the anomaly score $\mathcal{A} > \tau$, the image is classified as abnormal.

Specifically, Yu et al. (2021) propose FastFlow, which employs a lightweight 2D normalizing flow model to transform input image features into a standard normal distribution and uses probability values as anomaly scores to efficiently and accurately detect and localize anomalies. Rudolph et al. (2021) propose DifferNet, a semi-supervised defect detection method designed to address the challenge of defect detection caused by the scarcity and unknown characteristics of defect samples in the manufacturing process. DifferNet leverages a convolutional neural network for feature extraction and employs a normalizing flow to estimate feature density using multi-scale feature extraction and multi-transform scoring strategies, enabling high-precision defect detection and localization. With only a small number of training samples, DifferNet effectively detects and localizes defects, making it particularly suitable for real-world industrial applications where acquiring large defect datasets is challenging. Zhang et al. (2022) propose PEDENet, which integrates a block embedding network, a density estimation network, and a location prediction auxiliary network. Leveraging deep learning and density estimation methods, PEDENet accurately localizes anomalous regions in images using only normal images for training. To achieve precise anomaly localization, PEDENet employs a deep encoder for dimensionality reduction, a Gaussian mixture model to estimate the distribution of block embeddings, and a multilayer perceptron (MLP) to predict the relative positions of adjacent blocks. Gudovskiy et al. (2022) propose CFlow-AD, a real-time unsupervised anomaly detection and localization model designed for efficient anomaly detection in high-resolution images without labeled data or

anomalous samples. CFlow-AD is built upon a conditional normalizing flow framework and incorporates a pre-trained convolutional neural network for feature extraction. A multi-scale decoder estimates the likelihood of feature vectors, enabling fast and accurate anomaly detection and localization. Wan et al. (2022) propose position-enhanced feature mapping (PEFM), a novel image anomaly detection method designed to tackle the challenges of automated visual inspection in intelligent manufacturing systems. PEFM addresses detection performance and efficiency issues for various anomalies with different sizes, shapes, and colors. It integrates position encoding and feature mapping techniques to detect and localize anomalous regions. PEFM leverages pre-trained convolutional neural networks to extract and map image features. The mapping residuals between features extracted by different pre-trained convolutional neural networks are used as anomaly detection scores. Zheng et al. (2022) propose a coarse-to-fine non-contrastive learning method to address the challenge of detecting subtle and fine-grained anomalies in high-resolution images for industrial applications. This framework employs a coarse-to-fine alignment process. First, coarse alignment is performed at both the image and feature levels to standardize the pixel positions of objects. Then, during the fine alignment stage, a self-supervised learning approach-non-contrastive learning-is used to maximize the feature similarity at all corresponding positions within a batch, enabling the learning of a compact and semantically rich representation of normal images. Since this method does not require any prior assumptions about anomalous samples, it enhances the model's generalization across a wide range of anomalies and improves both anomaly detection accuracy and localization precision. Kamoona et al. (2024) propose a lightweight industrial defect detection method based on feature energy in the random finite set (RFS) framework. This method addresses the limitations of traditional global feature extraction techniques, which are not robust to lighting variations and viewing angle changes in manufacturing and fail to fully utilize geometric information. Kamoona et al. introduce transfer learning to process local features, modeling them as RFSs. They define RFS energy as the anomaly score and model the point pattern feature similarity distribution of normal samples using a multivariate Gaussian distribution. This approach enables anomaly detection parameter learning with low computational cost and significantly enhances anomaly detection accuracy. Rudolph et al. (2022) propose CS-Flow, a fully convolutional cross-scale normalization flow method designed for automatic defect detection in industrial manufacturing, particularly in scenarios where defect-free samples are unavailable. CS-Flow achieves high-precision defect detection at the image level by jointly processing feature maps at different scales and leveraging normalizing flow to flexibly estimate the density of input samples and assign probability values to them. Zhou et al. (2024) propose MSFlow, a multi-scale flow model framework designed to address the detection challenges posed by abnormal size variations in unsupervised anomaly detection. MSFlow enhances generalization ability to abnormal size changes by incorporating heterogeneous parallel flows and fusion flows, which capture features and facilitate multi-scale information exchange. MSFlow achieves significant performance improvements in both image-level and pixel-level anomaly detection and localization tasks. Cao et al. (2023) propose collaborative difference optimization (CDO) to mitigate the overgeneralization problem in unsupervised image anomaly detection, which arises due to the high generalization ability of convolutional neural networks. CDO leverages synthetic anomalies to refine feature difference distributions by optimizing the boundary and overlap between normal and anomalous samples. It incorporates a boundary optimization module and an overlap optimization module to

enhance the separation between normal and anomalous features, reducing feature overlap and improving both anomaly detection accuracy and localization reliability. To tackle the challenge of defect localization in industrial inspection caused by uncontrollable factors, Lei et al. (2023) propose PyramidFlow, which introduces a defect comparison and localization paradigm based on potential templates, combined with a pyramid-based normalizing flow for multi-scale fusion and volume normalization. Without relying on external priors, PyramidFlow effectively enhances model generalization while achieving high-resolution defect localization. Silva et al. (2024) propose AttentDifferNet, which enhances defect detection and classification in industrial inspection images by integrating an attention module into the DifferNet architecture. AttentDifferNet significantly improves model performance across multiple industrial inspection datasets by incorporating SENet and CBAM attention modules. Milkovic et al. (2024) propose FRAnomaly, a fast image anomaly detection method based on normalizing flows. FRAnomaly reduces model parameters and inference time by utilizing multiple feature sets at a single scale and a normalizing flow model specifically designed for these features. FRAnomaly effectively addresses the limitations of traditional anomaly detection methods, which suffer from insufficient performance and high model complexity, achieving efficient and accurate image anomaly detection.

2.2.4 Memory bank

Memory Bank, as an innovative long-term memory mechanism, has demonstrated unique advantages in industrial visual anomaly detection. By simulating human memory behavior, it provides a more intelligent and efficient solution for anomaly detection. Figure 6 illustrates the architecture of the memory bank-based approach. The core of memory bank lies in its adaptive memory updating mechanism. By recording and storing interaction data during industrial anomaly detection, it enables long-term retention and utilization of relevant information, thereby enhancing both detection accuracy and efficiency. Given an industrial image I , a pre-trained feature extraction network is first applied to map the image to feature space, obtaining a feature vector

$$f = E(I; \theta_E), \quad (20)$$

where E denotes the feature extraction network, and θ_E represents its parameters. The feature vector of the image is given by $f \in \mathbb{R}^d$, where d is the feature dimension. The memory

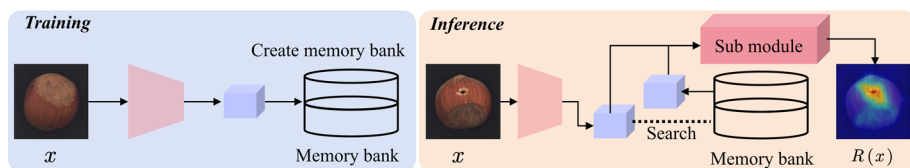


Fig. 6 Memory bank method. **Training phase:** The model extracts features from normal samples and stores these representations in a memory bank, thereby constructing a memory of the normal data distribution. **Inference phase:** Validation data undergo the same feature extraction process, and the resulting feature vectors are queried against the memory bank for the most similar entries. Anomaly score is then computed; if the discrepancy from the stored normal features is substantial, the input is deemed anomalous

bank stores a set of feature vectors from normal samples, denotes as $\mathcal{M}$, which is defined as follows:

$$\mathcal{M} = \{m_1, m_2, \dots, m_k\}, m_i \in \mathbb{R}^d. \quad (21)$$

where k denotes the size of the memory bank, i.e., the number of stored feature vectors from normal samples. m_i denotes the feature vector of a normal sample. During the inference stage, for a given input sample I , a feature vector f is extracted using the feature extraction network. The similarity or reconstruction error between f and all normal features $\mathcal{M}$ stored in the memory bank is then computed. To determine the nearest neighbor distance between the input feature f and the stored features m_i in the memory bank, either cosine similarity or Euclidean distance can be used.

$$S_i = \cos(f, m_i) = \frac{f \cdot m_i}{\|f\| \|m_i\|}. \quad (22)$$

$$D_i = \|f - m_i\|_2^2. \quad (23)$$

Select the minimum distance D_{min} or the maximum similarity S_{max} from the nearest neighbor:

$$D_{min} = \min_i D_i, S_{max} = \max_i S_i. \quad (24)$$

Define the anomaly score $\mathcal{A}$ based on the nearest neighbor distance or similarity as follows:

$$\mathcal{A} = D_{min} = \min_i \|f - m_i\|_2^2. \quad (25)$$

Based on cosine similarity, the anomaly score $\mathcal{A}$ is defined as:

$$\mathcal{A} = 1 - S_{max} = 1 - \max_i \cos(f, m_i). \quad (26)$$

Set a threshold ϵ . If the anomaly score $\mathcal{A} > \epsilon$, the input sample is classified as anomalous. To enhance detection performance, a reconstruction error mechanism can be introduced. The input feature f is mapped to the nearest neighbor feature space m_k and used to reconstruct the approximate feature $\hat{f}$:

$$\hat{f} = W(m_k), k = \arg \min_i \|f - m_i\|_2. \quad (27)$$

Where W denotes the reconstruction mapping function. The reconstruction error is defined as:

$$\mathcal{L}_{recon} = \|f - \hat{f}\|_2^2. \quad (28)$$

The final comprehensive anomaly score is given by:

$$\mathcal{A}_{recon} = \alpha \mathcal{A} + \beta \mathcal{L}_{recon}. \quad (29)$$

where α and β are weighting parameters.

In recent years, research on the application of memory bank in industrial visual anomaly detection has gradually gained attention. Researchers have proposed a series of unsupervised anomaly detection methods based on memory enhancement mechanisms and explored a memory-enhanced state-space model that integrates memory bank with a state-space framework. These studies aim to enhance the accuracy and efficiency of industrial visual anomaly detection and further drive the intelligentization and automation of industrial production. Defard et al. (2021) propose Patch Distribution Modeling (PaDiM), an industrial image anomaly detection and localization framework based on one-class learning. PaDiM employs a cosine-trained convolutional neural networks to extract embedding vectors from image patches and models the normal category distribution using a multivariate Gaussian distribution. Additionally, PaDiM captures correlations across different semantic levels of the CNN to enhance anomaly localization accuracy. Roth et al. (2022) propose PatchCore, which utilizes locally-aware, mid-level feature patches extracted from a pre-trained ImageNet model and employs a core-set reduction technique to construct a compact yet highly representative memory bank. This approach enables precise anomaly detection and localization while ensuring efficient inference speeds. Tsai et al. (2022) propose MSP, a multi-scale patch-based representation learning method designed to address the challenges of detecting small or complex-textured abnormal regions in image anomaly detection and segmentation. MSP enhances feature representation capability by sliding image patches at different scales to extract representative features, while simultaneously capturing both global and local contextual information. Additionally, MSP optimizes the loss function by incorporating K-Means clustering, cosine similarity, and self-supervised learning strategies, enabling the model to effectively learn feature representations from normal samples. Sungwook et al. (2022) propose coupled-hypersphere-based feature adaptation (CFA), an anomaly detection method designed to address the feature bias problem in traditional anomaly localization methods. CFA enhances anomaly detection and localization accuracy by adapting object dataset features through transfer learning, learnable patch descriptions, and an extensible memory bank that operates independently of the object dataset size. This enables CFA to accurately differentiate between normal and anomalous features. McIntosh and Albu (2023) propose InReaCh, an unsupervised anomaly detection method designed to overcome the limitations of traditional image anomaly detection, which relies on clean training data. InReaCh constructs channels by associating high-confidence conventional patches from the training data and develops a model based on these channels to evaluate new patches. Bae et al. (2023) propose PNI, an industrial anomaly detection and localization algorithm designed to address the lack of location and neighborhood information in industrial anomaly detection. PNI models the distribution of normal features by incorporating location and neighborhood information, utilizing a multilayer perceptron (MLP) network and histograms. Additionally, PNI introduces a pixel-level refinement network, trained on synthetic anomaly images, to enhance the detail and accuracy of the anomaly map. PNI significantly improves anomaly detection and localization accuracy, particularly in object-level anomaly classification. Liu et al. (2023) propose Diversity-Measurable Anomaly Detection, a novel anomaly detection framework designed to balance the trade-off between reconstructing diverse normal patterns and detecting unknown anomalies in reconstruction-based anomaly detection models. Diversity-measurable anomaly detection introduces a pyramid deformation module (PDM) to simulate and quantify the diversity among normal patterns and the

severity of anomalies. Additionally, it integrates PDM with an information compression module (ICM) to enable accurate reconstruction of diverse normal patterns and reliable severity measurement of anomalies without relying on external estimators. Kim et al. (2023) propose fast adaptive patch memory (FAPM), an industrial anomaly detection method designed to address the insufficient inference speed of existing approaches. FAPM leverages local and hierarchical memory banks to store embedded image features and employs local adaptive core-set sampling to eliminate redundant computations. This approach significantly enhances inference speed while preserving anomaly detection accuracy. Xie et al. (2023) propose GraphCore, a novel anomaly detection method that enables efficient anomaly detection with a limited number of samples. GraphCore leverages a graph neural network to extract visual isometric invariant features, enhancing detection performance under data-scarce conditions. To address the challenges in industrial product image defect detection, including the scarcity of defect samples and unclear classification standards, Yongheng et al. (2023) propose self-updating memory and central clustering (SMCC), a novel unsupervised anomaly detection and localization method. SMCC extracts image features using a pre-trained model and applies a Gaussian mixture model for clustering to obtain cluster centers, ensuring that normal sample features are tightly distributed around these centers to facilitate anomaly distinction. Additionally, SMCC incorporates a self-updating memory bank to reduce memory usage and optimizes the pre-trained network parameters to adapt to the current dataset distribution, thereby achieving efficient anomaly detection and localization while significantly enhancing accuracy and robustness. Hu et al. (2024) propose DMAD, an anomaly detection framework designed to address the multi-class anomaly detection problem in real-world industrial scenarios. DMAD enhances feature representation and learns anomaly scores by constructing a dual memory bank-comprising a normal memory bank and an anomaly memory bank-to compute feature distances and feature attentions. By integrating unsupervised and semi-supervised learning, DMAD significantly improves anomaly detection accuracy and robustness. Zhou et al. (2024) propose VQ-Flow, a method designed to mitigate performance degradation caused by pattern mixing among different categories in multi-class anomaly detection. VQ-Flow incorporates hierarchical vector quantization, enabling a normalizing flow model to differentiate between distinct data concepts without requiring category labels while capturing concept-specific normal patterns. VQ-Flow employs a conceptual prototype codebook to distinguish concepts and a concept-specific pattern codebook to learn concept-specific normal patterns. The model simulates complex multi-class data distributions by modeling conditional distributions in a concept-aware manner using a Gaussian mixture model. This approach facilitates effective modeling and anomaly detection in multi-class settings. Zuo et al. (2024) propose ReconFA, a self-supervised domain adaptation method designed to address mismatches between pre-trained convolution neural network features and the object data domain in anomaly detection. ReconFA effectively adapts pre-trained features to the object domain by introducing a self-supervised multi-scale feature fusion approach, which integrates a multi-scale aggregation module with a feature reconstruction task, thereby enhancing anomaly detection performance. To address the high computational complexity and feature mismatch issues in traditional 3D point cloud anomaly detection methods, Zhao et al. (2024) propose PointCore, an efficient unsupervised anomaly detection method for point clouds. PointCore integrates local coordinates and global PointMAE features through a single-repository design, assigns different priorities to these features, and introduces a normalized ranking method to

handle anomaly scores across different scales. PointCore reduces computational costs and interference during inference, thereby enhancing anomaly detection accuracy and efficiency. Shuai et al. (2024) propose reducing biases in representation (REB), an industrial anomaly detection method designed to address performance limitations caused by domain bias and local density bias in feature representations of pre-trained models. REB enhances anomaly detection accuracy by mitigating domain bias through self-supervised learning tasks and the DefectMaker defect generation strategy, while reducing local density bias using local-density KNN, which normalizes distances based on a local density model. Hyun et al. (2024) propose ReConPatch, a method designed to address the challenge of rare and unknown product defects in industrial manufacturing. ReConPatch utilizes contrastive representation learning to train a linearly modulated patch feature representation, incorporating pairwise and contextual similarities as pseudo labels. This enables the construction of a goal-oriented, easily separable feature space, thereby achieving high-performance industrial anomaly detection. Lee et al. (2022) propose CRAD, a novel anomaly detection method designed to address the poor generalization ability, identity shortcut, and high complexity issues in multi-class anomaly detection. CRAD enhances generalization ability and mitigates the identity shortcut problem by mapping spatial features onto coordinates in a continuous grid, fusing local and global normal features, and employing feature refinement techniques to achieve an effective representation of normal features in continuous space. Additionally, CRAD reduces storage and computational complexity. Koshil et al. (2024) propose AnomalousPatchCore (APC), which rely solely on normal samples and fail to fully utilize anomalous sample information. APC effectively leverages both normal and anomalous samples by fine-tuning a feature extractor and integrating three auxiliary tasks: classification, segmentation, and reconstruction. Wang et al. (2024) propose MemoryMamba, a memory-enhanced state-space model designed to overcome the limitations of existing defect recognition models in handling complex and limited defect data, particularly in imbalanced or data-scarce scenarios. MemoryMamba enhances defect recognition accuracy and robustness by integrating state-space techniques with memory-enhancement mechanisms. MemoryMamba employs coarse-grained and fine-grained memory encoding modules to capture and encode defect-hierarchy features while optimizing the memory network through contrastive learning and mutual information maximization strategies. He et al. (2024) propose MambaAD, which is designed to address the challenges of capturing long-range dependencies and the high computational complexity encountered by traditional convolution neural networks and transformers in anomaly detection. MambaAD employs a novel decoder built on the Mamba architecture, which integrates global and local feature extraction through local state-space augmentation modules and hybrid state-space (HSS) blocks.

2.2.5 Contrastive language-image pre-training

As demonstrated in Fig. 7, vision-language model-based industrial anomaly detection leverages visual features and linguistic descriptions to enhance anomaly detection accuracy through multimodal learning. These methods employ large-scale pre-trained multimodal models Radford et al. (2021) to jointly model image features and text descriptions, enabling visual anomaly identification. Although textual descriptions are less readily available in industrial settings than in general scenarios, multimodal approaches remain highly promising for industrial vision anomaly detection. In practice, textual information can be

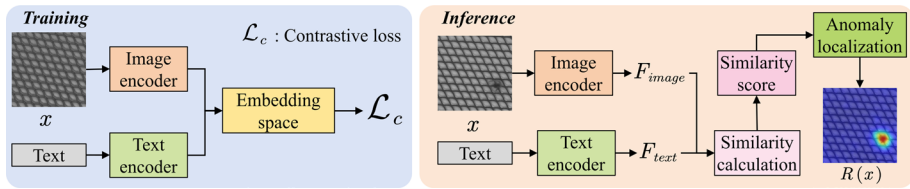


Fig. 7 Contrastive language-image pre-train (CLIP) method. **Training phase:** The CLIP model simultaneously takes an image x and its corresponding textual description as input. Through its image and text encoder, the image and text are mapped into a shared embedding space. A contrastive loss $\mathcal{L}_c$ is then applied to pull matched image-text pairs closer in this feature space and push mismatched pairs farther apart, thus achieving cross-modal alignment. **Inference phase:** Given a new image and text, the model's image and text encoder each generate feature representations (F_{image} and F_{text}), and a similarity score is computed. If the similarity score falls below a certain threshold, it indicates a significant discrepancy between the image and its textual description, and the input is flagged as anomalous

systematically extracted from production standards, defect taxonomies, and quality control guidelines, offering semantically rich references for anomalies that are otherwise difficult to distinguish visually. Moreover, integrating text facilitates more interpretable detection outcomes, enabling operators to better understand identified issues and take timely corrective actions. Thus, despite the relative scarcity of spontaneous textual annotations in industrial environments, multimodal methods that leverage structured, domain-specific textual resources remain both relevant and valuable for enhancing anomaly detection performance in real-world applications. Given an industrial image I , visual features are first extracted using a visual encoder:

$$F = E_{vis}(I). \quad (30)$$

where E_{vis} denotes the visual feature extractor. $F \in \mathbb{R}^d$ represents the global feature vector of the image, where d is the feature dimension. An image processed in blocks is also able to be represented as a sequence of features.

$$F = \{f_1, f_2, \dots, f_N\}. \quad (31)$$

A set of predefined text prompts $\mathcal{T} = \{T_1, T_2, \dots, T_k\}$ is used to describe normal or abnormal states. For example, T_1 : The object is normal; T_2 : The object is defective; T_3 : There are cracks on the surface. Text features are extracted using a language encoder:

$$L_j = E_{lang}(T_j), \quad j = 1, 2, \dots, k. \quad (32)$$

where E_{lang} denotes the language feature extractor. $L_j \in \mathbb{R}^d$ represents the embedding vector of the text description T_j . The core of the vision-language model is to compute the similarity between visual features F and text features L_j . This can be expressed using cosine similarity:

$$S_j = \cos(F, L_j) = \frac{F \cdot L_j}{\|F\| \|L_j\|}. \quad (33)$$

where, S_j represents the similarity between the input image I and the text description T_j . The similarity score $S_j \in [-1, 1]$, where a higher value indicates a better match between the visual features and the text description. For multiple text descriptions, the final similarity is expressed as a vector $S = \{S_1, S_2, \dots, S_k\}$. In industrial anomaly detection, it is assumed that normal samples exhibit high similarity to normal text descriptions, whereas abnormal samples align more closely with abnormal text descriptions. Based on this assumption, anomalies can be identified through similarity distribution analysis. The anomaly score $\mathcal{A}$ is defined as the difference in similarity between the normal and anomalous descriptions:

$$\mathcal{A} = \max_{j \in \mathcal{A}_T} S_j - \max_{j \in \mathcal{N}_T} S_j. \quad (34)$$

where $\mathcal{A}_T$ denotes the set of abnormal text descriptions. $\mathcal{N}_T$ denotes the set of normal text descriptions. $\max_{j \in \mathcal{A}_T} S_j$ represents the maximum similarity between the image and the abnormal text. $\max_{j \in \mathcal{N}_T} S_j$ represents the maximum similarity between the image and the normal text. To enhance the detection of local anomalies, the visual features of local image patches can be matched with linguistic descriptions to compute the similarity of each local region. Given an image block feature F_i , the local similarity is computed as:

$$S_{i,j} = \cos(F_i, L_j). \quad (35)$$

The local anomaly score $\mathcal{A}_{local}$ is defined for all image patches as:

$$\mathcal{A}_{local} = \max_{i,j \in \mathcal{A}_T} S_{i,j} - \max_{i,j \in \mathcal{N}_T} S_{i,j}. \quad (36)$$

By combining the global and local anomaly scores, the final anomaly score $\mathcal{A}_{final}$ is defined as:

$$\mathcal{A}_{final} = \alpha \mathcal{A} + \beta \mathcal{A}_{local}, \quad (37)$$

where α and β are weighting parameters.

Contrastive learning has recently emerged as a powerful approach in industrial anomaly detection, particularly through vision-language alignment. Unlike traditional self-supervised learning methods such as image reconstruction, pseudo-anomaly generation, or masked modeling, contrastive learning builds discriminative embedding spaces by constructing positive and negative pairs. Models like CLIP and WinCLIP leverage multimodal supervision, enabling zero-shot anomaly detection and demonstrating strong generalization across object categories and domain shifts. On the other hand, traditional self-supervised methods focus on modeling intra-image structure or reconstructing visual content to detect anomalies. These methods often perform well in single-modal, structured environments and are well-suited to tasks with clearly defined defect types.

Recently, researchers have started to explore how contrastive learning-particularly vision-language models-can be adapted to industrial contexts to support semantic anomaly detection and localization. One representative work in this direction is WinCLIP, which builds on the CLIP foundation and introduces mechanisms specifically tailored to anomaly segmentation tasks. Specifically, Jeong et al. (2023) propose WinCLIP, which is built upon the vision-language model CLIP and achieves zero-shot anomaly segmentation aligned with

text by integrating compositional prompting and multi-scale feature extraction and aggregation. Additionally, Jeong et al. introduce WinCLIP+, an enhanced version that further improves classification and segmentation accuracy by incorporating additional information from normal images. Tamura (2023) propose RWDA, which requires large numbers of anomaly samples and a separate model for each object category. RWDA leverages the CLIP as a data source and applies random data augmentation to generate diverse text embeddings as training data. These embeddings are then used to train a feedforward neural network, enabling the classification of normal and abnormal samples without requiring category-specific training images or a cumbersome prompt integration process. Chen et al. (2023) propose CLIP-AD, a zero-shot anomaly detection framework designed to address the challenge of detecting anomalies without reference images for the test object. CLIP-AD leverages the zero-shot capabilities of the large-scale CLIP and enhances text prompt design through the representative vector selection paradigm. Additionally, it introduces a structured feature modification strategy to construct a staged dual-path (SDP) model for handling abnormality graph computation. To further address feature misalignment, a fine-tuned linear layer (SDP+) is incorporated, enabling efficient zero-shot anomaly detection. To tackle the challenge of unified zero-shot anomaly detection and localization, Deng et al. (2023) propose AnoCLIP, which extracts local features from the CLIP model by introducing a train-free value-to-value attention mechanism and designs a domain-aware contrastive state prompt template to enhance vision-language alignment. Additionally, AnoCLIP incorporates a fast test-time adaptation mechanism that optimizes the alignment between visual markers and text prompts, significantly improving the accuracy of anomaly detection and localization. Zhou et al. (2024) propose AnomalyCLIP, an industrial anomaly detection method designed to address the challenge of model generalization to diverse domain-specific anomaly patterns in zero-shot anomaly detection through text prompt learning without object semantics. AnomalyCLIP learns object-agnostic text prompts and leverages the large-scale pre-trained CLIP to capture general normalcy and abnormality in images. This enables the model to focus on anomalous regions rather than object semantics, thereby achieving generalized anomaly recognition across various object categories. Additionally, AnomalyCLIP incorporates global and local context optimization as well as text and visual space alignment, significantly enhancing anomaly detection and segmentation performance across diverse datasets. Cai et al. (2024) propose CLIP-ADA, an anomaly detection framework designed to address the challenges of high resource consumption and low localization accuracy in multi-class industrial anomaly detection. This is achieved by introducing learnable prompts and an anomaly region refinement strategy. CLIP-ADA incorporates learnable textual prompts and associates them with anomalous patterns through self-supervised learning, enabling unified anomaly detection across multiple categories. Additionally, CLIP-ADA introduces a coarse-to-fine anomaly region refinement strategy, utilizing the initial coarse localization results as an attention map to enhance the model's ability to accurately identify and localize anomaly regions. Li et al. (2024) propose PromptAD, an industrial anomaly detection method designed to address the limitations of traditional anomaly detection approaches, which rely heavily on manually designed prompts and lack automation capabilities. PromptAD leverages semantic concatenation to transform normal sample prompts into abnormal sample prompts and constructs a large number of negative samples to facilitate prompt learning in single-class settings. Additionally, PromptAD introduces the concept of explicit anomaly margin, which explicitly regulates the distance between normal and abnormal prompt fea-

tures through hyperparameters, thereby enabling effective anomaly detection using only normal samples. To accurately detect and localize anomalies in industrial products without a priori samples for the target category, Gu et al. (2024) propose a zero-shot anomaly detection method called FiLo, which leverages fine-grained descriptions from adaptive learning and location-enhanced high-quality localization. It utilizes a large language model to generate detailed anomaly descriptions and incorporates a multi-scale, multi-shape cross-modal interaction module to enhance the localization capability for anomalies of various sizes and shapes. These enhancements significantly improve the accuracy and interpretability of zero-shot anomaly detection and localization. To address the challenge of accurately identifying and localizing anomalous patterns in visual data without training samples from the target dataset, Zhang et al. (2024) propose dual-image enhanced CLIP (DiE-CLIP), which significantly improves anomaly classification and localization performance by utilizing paired images as visual references and integrating the joint visual-linguistic scoring system of the CLIP model. Additionally, DiE-CLIP introduces a test-time adaptation model, which further enhances localization capability by synthesizing anomalies, thereby achieving substantial improvements in zero-shot anomaly detection and localization. Lee et al. (2024) propose AnoPLe, a multimodal prompting learning method designed to address the challenge of anomalous sample scarcity in small-sample anomaly detection. AnoPLe detects anomalies through bidirectional prompting learning and multimodal interaction. It leverages simulated anomalies and lightweight decoders to enhance local semantic understanding, while improving image-level anomaly detection performance by integrating global and local semantics. Li et al. (2024) propose a zero-shot industrial anomaly detection framework IAD-CLIP based on visual language models. The framework contains a pre-trained CLIP model, a training-free adaptation module and a test-time adaptation mechanism. The training-free adaptation module uses a value-value attention mechanism and a state prompt space. The pre-trained CLIP model is used for feature extraction and the training-free adaptation module processes the extracted features through visual coders and text encoders for anomaly detection and localization.

2.2.6 Anomaly clustering

Wan et al. (2021) propose Gaussian clustering pre-trained features (GCPF), a framework designed to address the computational complexity and memory consumption challenges in industrial anomaly localization. GCPF extracts features using a pre-trained convolutional neural network and achieves pixel-level anomaly localization through multi-independent multivariate Gaussian clustering (MIMGC) and multi-level anomaly scoring. By significantly reducing computational time and memory storage requirements while improving anomaly localization accuracy, GCPF enhances practical applicability in industrial settings. Sohn et al. (2023) propose a framework that leverages pre-trained deep embeddings and off-the-shelf clustering methods to address the challenge of clustering different types of anomalies in abnormal images. The framework measures image similarity by defining a distance function based on patch embeddings, where the similarity weights are computed by weighted averaging the Euclidean distances of the embeddings. These weights are estimated in an unsupervised or semi-supervised manner to effectively highlight defective regions. Zhang et al. (2023) propose JRCC-Net, a segmentation network designed for unsupervised surface anomaly detection. JRCC-Net learns the decision boundary by leveraging

the distance difference between simulated abnormal and normal samples, incorporating a simulated anomaly strategy, a joint representation, and a contrastive clustering method to enable anomaly detection during inference. Yao et al. (2024) propose hierarchical Gaussian mixture normal flow (HGAD), a modeling method designed to address the uniform mapping problem in normalizing flow-based anomaly detection methods for unified anomaly detection tasks. HGAD enhances the model's ability to learn multi-class normal data distributions by integrating inter-class Gaussian mixture modeling and intra-class mixed class center learning, preventing performance degradation caused by mapping different classes to the same Gaussian prior. Additionally, HGAD introduces mutual information maximization loss to improve class distinction and enhance the model's discriminative capability.

2.3 Reconstruction-Based Methods

2.3.1 Autoencoder

Autoencoder is an unsupervised learning neural network model that compresses input data into a low-dimensional representation through an encoder-decoder structure and then reconstructs an output similar to the original data. Figure 8 illustrates the fundamental architecture of the reconstruction-based method. The objective is to minimize reconstruction error, thereby achieving an effective data representation. An Autoencoder consists of two main components: an encoder and a decoder. The encoder compresses the input data into a low-dimensional representation, while the decoder reconstructs it to approximate the original data as closely as possible. Given an input image I , an Autoencoder can be formulated as:

$$I \xrightarrow{\text{Encoder}} z \xrightarrow{\text{Decoder}} \hat{I}, \quad (38)$$

where z represents the low-dimension representation, and $\hat{I}$ is the image reconstructed by the decoder. The encoder aims to minimize the difference between the reconstructed image and the original image. The MSE is commonly used to quantify this difference between the original image I and the reconstructed image $\hat{I}$:

$$\mathcal{L}(I, \hat{I}) = \|I - \hat{I}\|_2^2 = \sum_{i=1}^n (I_i - \hat{I}_i)^2, \quad (39)$$

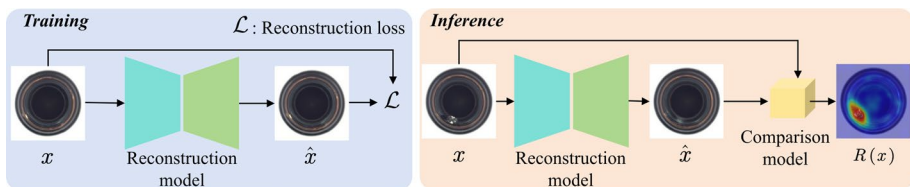


Fig. 8 Reconstruction method. **Training phase:** The model learns to reconstruct the primary features of normal images x . Specifically, x is passed through the reconstruction network to produce $\hat{x}$, and a reconstruction loss $\mathcal{L}$ measures the discrepancy between $\hat{x}$ and x . By iteratively minimizing this loss, the model refines its ability to capture the intrinsic structure and distribution of normal data. **Inference phase:** Given a new input image x , the trained reconstruction model generates $\hat{x}$, after which both x and $\hat{x}$ are fed into a comparison module to assess their differences. A substantial mismatch indicates the model's inability to faithfully reconstruct the input, often signaling anomalous regions within the image

where both X and $\hat{X}$ are n -dimensional vectors representing the pixel values of the image.

In industrial anomaly detection, the assumption is that normal images are well reconstructed by the autoencoder, whereas abnormal images show higher reconstruction errors. Thus, the reconstruction error serves as a criterion for determining whether an image is anomalous. A threshold ϵ is set, and if the reconstruction error exceeds this threshold, the image is classified as anomalous:

$$\begin{cases} \mathcal{L}(I, \hat{I}) > \epsilon, \mathbf{I} \text{ is abnormal} \\ \mathcal{L}(I, \hat{I}) \leq \epsilon, \mathbf{I} \text{ is normal} \end{cases} \quad (40)$$

During the training phase, the autoencoder's parameters are optimized by minimizing the reconstruction error. A commonly used optimization method is gradient descent, with the update rule given by:

$$\theta^* = \arg \min_{\theta} \sum_{i=1}^m \mathcal{L}(I_i, \hat{I}_i; \theta), \quad (41)$$

where θ represents the parameter of the self-encoder, I_i is the i -th image in the training set, and m is the number of training samples.

In industrial anomaly detection, Autoencoders learn normal data representations and identify anomalous samples that significantly deviate from them. Their advantages include unsupervised learning, effective dimensionality reduction, feature extraction, and the ability to capture complex nonlinear relationships. However, a key limitation is that an overly complex model or an excessively large bottleneck layer may reconstruct the input too accurately, failing to capture critical features. To address these issues, researchers have proposed various Autoencoder variants, such as sparse Autoencoders, denoising Autoencoders, and variational Autoencoders Nguyen et al. (2024). These variants improve the performance and applicability of Autoencoders by introducing additional constraints and optimized objective functions. Zavrtanik et al. (2021) propose DRAEM, a surface anomaly detection method designed to identify local regions in an image that significantly deviates from normal appearance. DRAEM integrates a reconstruction subnetwork and a discriminative subnetwork to jointly learn representations of abnormal images and their normal reconstructions, while simultaneously defining the decision boundary between normal and abnormal samples. This enables direct anomaly localization, allowing for accurate detection and segmentation of anomalous regions without requiring complex post-processing steps. Zavrtanik et al. (2021) propose RIAD, a visual anomaly detection method based on image restoration. RIAD detects anomalies by segmenting an image, randomly removing certain regions, and then reconstructing these regions using a trained encoder-decoder network. The key idea behind RIAD is that abnormal regions are difficult to reconstruct accurately because they were not observed during training, whereas normal regions are well reconstructed. By comparing the differences between the input image and the reconstructed image, RIAD effectively detects and localizes anomalies. Additionally, RIAD introduces a gradient similarity-based loss function to train the network, significantly improving visual anomaly detection performance. Shi et al. (2021) propose DFR, an unsupervised anomaly segmentation method based on deep feature reconstruction, designed to automatically detect and

localize small, hard-to-detect anomaly regions in images without prior knowledge of the anomalies. DFR leverages a pre-trained deep convolutional network to extract multi-scale regional features and employs an efficient convolutional autoencoder to rapidly identify anomalies through feature reconstruction. Hou et al. (2021) propose divide-and-assemble anomaly detection (DAAD), a framework that formulates image reconstruction as a “divide-and-assemble” process. DAAD introduces a multi-scale block-level memory module to regulate the reconstruction ability for normal and abnormal samples, while employing adversarial learning to enhance semantic feature representation, enabling the detection of anomalies in both image space and semantic feature space. DAAD effectively addresses the challenge of distinguishing normal and abnormal images in unsupervised anomaly detection scenarios where anomalous samples are unavailable. Yang et al. (2022) propose DFC, an unsupervised anomaly detection and segmentation method based on deep feature correspondence. DFC addresses the challenge of effectively detecting and segmenting abnormal structures in images when anomalous samples are scarce or unavailable. DFC constructs an asymmetric dual-network framework, comprising a feature extraction network and a feature estimation network. DFC employs a deep feature correspondence mechanism to model and evaluate feature consistency between the two network branches, thereby enabling anomaly detection and segmentation. Additionally, DFC introduces a self-feature enhancement strategy and a multi-context residual learning module, significantly enhancing robustness and detection performance. Zavrtnik et al. (2022) propose DSR, a novel unsupervised surface anomaly detection framework designed to overcome the limitations of existing methods in synthesizing anomalies that closely resemble normal patterns. Instead of relying on traditional image synthesis-based anomaly generation, DSR employs a quantized feature space representation and introduces two decoders to reconstruct general object appearance and specific object appearance separately, thereby achieving more robust anomaly detection and localization. Schlüter et al. (2022) propose natural synthetic anomalies (NSA), a self-supervised learning method designed for anomaly detection and localization using only normal training data. NSA generates natural and diverse synthetic anomalies by seamlessly blending scaled blocks from different images into the target image using Poisson image editing techniques. This approach eliminates the obvious discontinuities present in existing methods, thereby significantly enhancing anomaly detection and localization performance. To enhance the reconstruction and identification of anomalous samples in anomaly detection, Ristea et al. (2022) propose SSPCAB, a self-supervised pre-trained convolutional attention block. SSPCAB trains the model without requiring anomalous samples by predicting masked or missing information in the input data and using reconstruction error as an anomaly score. This approach significantly improves anomaly detection performance. To address the challenges of small sample availability and cross-product category adaptation in industrial anomaly detection, Fang et al. (2023) propose FastRecon, a method designed for rapid anomaly detection in new products under scarce data conditions. FastRecon employs feature reconstruction, leveraging a small number of normal samples to generate a normal version of the query sample. It then applies a regression algorithm with distribution regularization to determine the optimal transformation, ensuring that the reconstructed output remains both visually similar to the query sample and consistent with normal sample characteristics. Guo et al. (2023) propose THFR, a template-guided hierarchical feature restoration method designed to detect abnormalities of various sizes in complex normal patterns. THFR preserves normal features and filters anomalous features by introducing bottleneck

compression. It then restores distorted features through a template-guided compensation mechanism, converting anomalous features into normal representations. THFR enables accurate detection and localization of abnormalities across different scales. Madan and Ristea (2023) propose SSMCTB, an anomaly detection method that masks input information at any layer of the neural network and reconstructs the masked data using 3D convolutions and transformer blocks. SSMCTB is trained exclusively on normal samples, eliminating the need for anomalous data. Park et al. (2024) propose loss amplification (LAMP), a method that enhances anomaly detection performance by modifying the loss function without altering the neural network architecture. LAMP steepens the loss landscape, leading to larger reconstruction errors for unseen anomalous patterns, thereby improving detection in unsupervised anomaly detection settings. Chen et al. (2024) propose implicit foreground-guided network (IFgNet) to mitigate background noise misclassification in industrial anomaly detection. IFgNet adopts a multi-task learning framework, integrating foreground detection and image reconstruction tasks. By employing shared networks and task-specific attention mechanisms, IFgNet learns to distinguish foreground from background, thereby reducing background interference and enhancing anomaly detection accuracy. Liu and Wang (2024) propose multi-category anomaly autoencoder (MAAE), a method designed to address high storage costs, low training efficiency, and performance degradation in multi-category industrial anomaly detection. MAAE captures global category information and feature distributions through spatial and channel attention mechanisms. Additionally, it introduces an adaptive noise generator and a feature fusion module to simulate real noise and preserve surface semantics, enabling unified anomaly detection across different categories while enhancing model performance and generalization ability. Fan et al. (2024) propose PatchAnomaly, a patch-level data augmentation method that reformulates unsupervised anomaly detection as a supervised object detection task. PatchAnomaly synthesizes similar anomalies through self-supervised learning, enabling direct anomaly localization without the need for complex post-processing steps. Zhang et al. (2024) propose DC-AE, an unsupervised anomaly detection framework designed to address the issue of traditional autoencoders mistakenly reconstructing anomalies as normal features. DC-AE enhances anomaly detection accuracy by integrating a double-constrained autoencoder (DCAE) and an adaptive weighted similarity space attention (AWSSA) mechanism. It employs adversarial learning and a global memory bank to restrict the model's generalization ability, preventing incorrect reconstruction of anomalous regions. Additionally, AWSSA strengthens the model's focus on anomalies, further improving anomaly localization accuracy.

2.3.2 Generative adversarial networks

Generative adversarial networks show great potential in industrial anomaly detection. By training a model with normal samples and using the difference in feature distributions between normal and abnormal samples to identify anomalies, GANs effectively solve the problem of the scarcity of abnormal samples in industrial scenarios. In addition, generative adversarial networks can generate high-quality simulated data, which is helpful for training and testing anomaly detection models in the absence of actual abnormal data Xia et al. (2022). The goal of a GAN is to train the generator G to produce samples that closely resemble the true data distribution $p_{data}(x)$, while the discriminator D aims to distinguish between real and generated samples. The overall optimization objective of GAN is:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{data}} [\log D(x)] + \mathbb{E}_{z \sim p_z} [\log(1 - D(G(z)))], \quad (42)$$

where x represents a real sample, $z \sim p_z$ is a random noise input to the generator, $G(z)$ denotes the sample generated by the generator, $D(x)$ is the discriminator's output for the real sample, and $D(G(z))$ is the discriminator's output for the generated sample. The input image I is reconstructed by the generator G , producing $\hat{I}$. The reconstruction error $\mathcal{L}_{recon}$ serves as an indicator for anomaly detection:

$$\mathcal{L}_{recon} = \|I - G(I)\|^2, \quad (43)$$

The discriminator's output D for the input I and the generated sample $\hat{I}$ provides an additional basis for anomaly detection. The discriminator loss $\mathcal{L}_{disc}$ is:

$$\mathcal{L}_{disc} = -\log D(I) - \log(1 - D(G(I))), \quad (44)$$

To enhance detection accuracy, the reconstruction error and discriminator loss are typically combined to formulate a comprehensive anomaly score $\mathcal{A}(I)$:

$$\mathcal{A}(I) = \alpha \mathcal{L}_{recon} + \beta \mathcal{L}_{disc}, \quad (45)$$

where α and β are weight parameters that balance the contributions of the reconstruction error and the discriminator loss. An anomaly is detected when $\mathcal{A}(I)$ exceeds the predefined threshold ϵ .

Although GANs have demonstrated strong performance in industrial anomaly detection, they still face several critical challenges. First, the training process of GANs can be unstable and is often prone to issues such as mode collapse. Second, industrial data is typically noisy and contains diverse types of anomalies, which requires GANs to exhibit strong robustness against both noise and varying defect patterns. As one of the earliest approaches used for synthetic anomaly generation, GAN-based methods learn to produce defective patterns by perturbing or reconstructing normal images, thereby enriching the training distribution with pseudo-anomalies and enhancing the model's ability to identify defects. For instance, Yan et al. (2021) propose a semantic context-based learning method, SCADN, to address the challenge of detecting abnormal samples with low probability density in unsupervised anomaly detection datasets. SCADN generates multi-scale striped masks to obscure the semantic context of normal samples. During the inference phase, an error map is constructed by computing the differences between the reconstructed and input images, facilitating the identification of abnormal samples. This approach effectively mitigates the limitations of traditional anomaly detection methods in complex scenarios. Liang et al. (2023) introduce OCR-GAN, a novel approach to mitigating performance degradation caused by the limited reconstruction capability of traditional anomaly detection methods. OCR-GAN significantly improves reconstruction-based unsupervised anomaly detection by integrating an innovative frequency decoupling module to decompose the input image into different frequency distributions and a channel selection module to facilitate information interaction among multiple frequency branches. Duan et al. (2023) propose DFMGAN, a method for generating defective images with limited samples, to address the challenge of insufficient defect detection performance in industrial manufacturing due to the scarcity of real

defective images. DFMGAN employs a two-stage training strategy: first, the StyleGAN2 backbone network is trained on a large set of defect-free images. Then, a defect perception residual block is incorporated to generate realistic defect masks. These masks are used to manipulate features within the masked regions, enabling the generation of high-quality, diverse defective images. This approach effectively augments the dataset and enhances defect detection performance.

2.3.3 Transformer

In recent years, YOLO-based models such as YOLOv5 and YOLOv8 have been widely used in industrial visual anomaly detection tasks due to their efficient inference speed and lightweight architecture. These methods are based on convolutional neural networks and are good at quickly locating and identifying defects with obvious features in images, such as scratches, breaks, or abnormal contours. They are suitable for production environments that require high real-time performance and deployment efficiency. In scenarios where the boundaries of abnormal areas are clear and there is sufficient labeled data, the YOLO method can achieve high-precision detection with low latency. In contrast, Transformer-based methods use self-attention mechanisms to construct global context relationships, which have stronger feature expression capabilities and can effectively capture long-distance dependencies and fine-grained differences in images. They are especially suitable for processing situations where abnormal patterns are fuzzy, the background is complex, or abnormal signals are embedded in local textures. These methods show higher robustness and generalization ability when dealing with uneven lighting, background interference, or irregular structures in industrial environments. Although their computational resource consumption is relatively high, with the development of lightweight Transformer architectures, their disadvantage in real-time performance is gradually being reduced.

Specifically, Mishra et al. (2021) propose VT-ADL, an image anomaly detection and localization method based on the vision transformer, designed to automatically identify significant changes in industrial images. VT-ADL leverages reconstruction and patch embedding techniques to preserve spatial information within the Transformer network and integrates a Gaussian mixture density network to localize anomalies. You et al. (2022) propose ADTR, an anomaly detection method designed to distinguish and detect anomalies using only normal samples. Instead of directly processing raw pixel values, ADTR employs a transformer model to reconstruct pre-trained deep features and optimizes the model with a novel loss function, enabling effective anomaly detection even in the absence of abnormal samples. To overcome the limitations of existing anomaly detection methods in capturing global information, Lee and Kang (2022) propose AnoViT, which leverages global relationships, addressing the shortcomings of traditional convolutional neural networks, which are limited to local feature extraction. By utilizing the self-attention mechanism of ViT, AnoViT captures global contextual information and integrates it with convolutional layers to preserve spatial details, enabling accurate anomaly detection and localization. De et al. (2022) propose MeTAL, a masked transformer model based on the vision transformer architecture, designed to address the limitations of existing image anomaly detection methods in reconstructing and identifying anomalous regions. MeTAL employs a masked multi-head self-attention mechanism to reconstruct each image block using only surrounding block information, effectively disregarding potential anomalies within the block itself. Addition-

ally, MeTAL introduces a multi-shaped block structure, incorporating horizontal and vertical striped blocks alongside traditional square blocks. Their encodings are then combined to form the final block representation. Compared to using only square blocks, this multi-shaped block approach significantly enhances model performance. Pirnay and Chai (2022) propose inpainting transformer (InTra), a novel anomaly detection method designed to address the misdetection problem caused by the ability of traditional convolutional autoencoders to reconstruct anomalous regions. InTra leverages a transformer model based on self-attention mechanisms to reconstruct masked patches, enabling the repair and reconstruction of image blocks without relying on convolution operations. By incorporating global contextual information, InTra achieves effective detection and localization of anomalous regions. Jiang et al. (2022) propose masked swin transformer UNet (MSTUNet), an industrial anomaly detection model designed to address the limitations of traditional convolutional neural networks in extracting global contextual information. MSTUNet applies anomaly simulation and masking strategies to anomaly-free samples to generate simulated anomalies and leverages the swin transformer's powerful global learning capability to reconstruct masked regions. Meanwhile, it incorporates a UNet network for end-to-end anomaly detection, enabling high-precision industrial anomaly detection and localization. Zhang et al. (2025) propose a novel symmetric structure ViTAD based on the pure vision transformer (ViT) for the multi-class unsupervised anomaly detection (MUAD) task. In contrast to existing methods that rely on complex pyramid structures and a large number of manual designs, ViTAD adopts a simple pure ViT architecture, and through macro and micro design optimizations, it achieves efficient and powerful anomaly detection capabilities. In terms of macroscopic design, ViTAD removes the jump connection of multi-scale features and only uses the features of the last stage as input, avoiding potential information leakage caused by too short information flow paths. In terms of microscopic design, ViTAD carries out meticulous structural optimization. It uses features before normalization as fusion features, retains position embedding to enhance feature expression ability, and removes category labels to avoid interference with spatial structure. Yao et al. (2023) propose focus-the-discrepancy (FOD), an image anomaly detection framework designed to overcome the limitations of traditional methods in detecting complex anomalies due to insufficient analysis of local feature differences and internal-external image correlations. FOD leverages the transformer's self-attention mechanism to establish correlations within and between images. Additionally, it employs self-supervised learning with object relevance based on the RBF kernel, along with object relevance and entropy constraint strategies, to enhance the distinction between normal and abnormal patterns, enabling effective anomaly detection and localization. Luo et al. (2024) propose AMI-Net, an adaptive mask-filling network designed to address the limitations of unsupervised reconstruction methods in industrial anomaly detection, which struggle to effectively distinguish normal and abnormal regions in defect detection. AMI-Net enhances the accuracy and real-time performance of anomaly detection by introducing an adaptive mask generator, integrating multi-scale pre-trained feature extraction, and employing a random position quantitative masking strategy. During the testing phase, AMI-Net dynamically masks defect regions while preserving normal regions using the adaptive mask generator. Yao et al. (2024) propose prior normality prompt transformer (PNPT), an industrial anomaly detection method designed to address the same mapping problem encountered by traditional methods in multi-class detection. PNPT reconstructs the inherent characteristics of samples by introducing normality prompting and a dual-stream model structure to

capture category-specific normality features. Its innovative architecture comprises four key modules—class-specific normality prompting pool, hierarchical patch embedding, semantic alignment coupling encoding, and contextual semantic conditional decoding—which effectively distinguish normal from abnormal instances, significantly enhancing multi-class industrial image anomaly detection performance.

2.3.4 Diffusion model

In industrial anomaly detection, the diffusion model reconstructs high-quality images through an inverse process that removes abnormal regions from the input image Hao et al. (2023). Anomalies are detected by pixel-by-pixel and feature-by-feature comparisons between the input and reconstructed images. Additionally, the conditional denoising process ensures consistent recovery of the object image, enhancing anomaly detection accuracy. The diffusion model localizes anomalies through pixel-level and feature-level comparisons. Pixel-level comparisons analyze differences at the individual pixel level, while feature-level comparisons capture high-level semantic variations. Anomaly scores are computed based on the discrepancies between the input and reconstructed images, quantifying the severity of anomalies.

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I), \quad (46)$$

where x_t denotes the noisy data at step t , $\beta_t \in (0, 1)$ is a predefined parameter controlling the noise increment. After T steps, the distribution of X_T follows $X_T \sim \mathcal{N}(0, I)$. By the property of closure, the data distribution at any step t can be explicitly expressed as:

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I), \quad (47)$$

where $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$ denotes the cumulative noise attenuation factor. The objective of the denoising process is to reconstruct the original data x_0 in reverse by learning the denoising distribution $p_\theta(x_{t-1}|x_t)$ and progressively removing noise.

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sum_\theta(x_t, t)), \quad (48)$$

where $\mu_\theta(x_t, t)$ denotes the mean function learned by the model, and $\sum_\theta(x_t, t)$ represents the learned variance function. The objective is to minimize the denoising error, enabling the model to accurately reconstruct the original data from noise. The training objective of the diffusion model is to minimize the denoising error, which is typically optimized using the following objective function:

$$\mathcal{L}_{\text{denoise}} = \mathbb{E}_{x_0, t, \epsilon} [\|\epsilon - \epsilon_\theta(x_t, t)\|^2], \quad (49)$$

where $x_0 \sim q(x_0)$ denotes a real sample, $t \sim \text{Uniform}(1, T)$ represents the diffusion step, and $\epsilon \sim \mathcal{N}(0, I)$ is Gaussian noise.

Specifically, Wyatt et al. (2022) propose AnoDDPM, an anomaly detection method based on denoising diffusion probabilistic models (DDPMs). AnoDDPM addresses the challenge of detecting large abnormal regions in high-resolution images by replacing traditional

Gaussian noise with multi-scale simplex noise, effectively enhancing anomaly identification and localization. Zhang et al. (2023) propose DiffusionAD, an industrial anomaly detection method designed to overcome the performance limitations of traditional generative models caused by poor reconstruction quality. DiffusionAD enhances anomaly detection and localization accuracy by leveraging a denoising diffusion model and introducing a noise-to-norm paradigm and a fast one-step denoising paradigm, significantly improving both efficiency and precision. Mousakhan et al. (2023) propose Denoising Diffusion Anomaly Detection (DDAD), a novel anomaly detection method designed to overcome the limitations of traditional reconstruction-based approaches. DDAD guides the image denoising process through a conditional mechanism, integrates pixel-level and feature-level comparisons for anomaly localization, and employs domain adaptation techniques to enhance the feature extractor, significantly improving anomaly detection and localization performance. Yin et al. (2023) propose LafitE, an unsupervised multi-class anomaly detection method designed to address the “identity shortcut” problem in anomaly detection. LafitE leverages a latent diffusion model to reconstruct the normal feature distribution and incorporates a feature editing strategy to enhance reconstruction quality and reduce false positives. Additionally, LafitE introduces a hyperparameter selection strategy that constructs a pseudo-validation set by synthesizing anomalous data, significantly improving anomaly detection and localization accuracy. Zhang et al. (2023) propose DiffAD, an unsupervised surface anomaly detection method designed to address the challenges of structural changes and reconstruction difficulties in abnormal regions faced by traditional reconstruction-based methods. DiffAD leverages a latent diffusion model to generate high-quality, diverse images and incorporates noise conditional embedding and interpolation channel techniques to enhance anomaly differentiation. By reducing pixel-level differences between reconstructed and original images, DiffAD significantly improves anomaly detection and localization accuracy. Lu et al. (2023) propose RAN, an unsupervised anomaly detection method based on a denoising diffusion model. RAN introduces random noise to mask abnormal pixels and leverages the intermediate denoising process of the generative diffusion model to compute pixel-level anomaly scores, using KL divergence to enhance detection accuracy. Additionally, RAN incorporates a gradient denoising process to smoothly transform anomalous images into normal ones, addressing the limitation of traditional reconstruction methods, where abnormal regions may be inadvertently reconstructed, leading to poor detection performance. He et al. (2024) propose DiAD, a diffusion-based multi-class anomaly detection framework designed to preserve image category and pixel-level structural integrity in anomaly detection. DiAD enhances reconstruction accuracy by integrating a spatial-aware feature fusion block with a pixel-space autoencoder, a semantic guidance network in latent space, and a pre-trained feature extractor in feature space. By reconstructing anomalous regions while preserving the semantic integrity of the original image, DiAD significantly improves anomaly detection and localization performance. Teng et al. (2024) propose AnomalyDiffusion, a method for generating anomalous images from limited samples, designed to address the scarcity of anomalous data in industrial anomaly detection. AnomalyDiffusion leverages the latent diffusion model’s powerful prior information, integrating spatial anomaly embedding and an adaptive attention reweighting mechanism to generate high-quality anomalous images with precise anomaly mask alignment. Tebbe and Tayyub (2024) propose dynamic denoising diffusion probability model (D3AD) to address the challenge of precisely localizing anomalies of different scales in anomaly detection. D3AD enhances localization accuracy

by introducing a dynamically calculated noise step, leveraging latent space representations, and employing a domain-adaptive feature extractor. Specifically, D3AD incorporates a dynamic implicit conditioning mechanism that adjusts noise addition based on the initial anomaly prediction, enabling more precise anomaly segmentation. Additionally, it performs denoising in latent space rather than at the pixel level, reducing computational demands and improving reconstruction quality for large missing components. By accurately determining the optimal number of noise steps, D3AD significantly improves anomaly detection performance, particularly in anomaly localization. Yao et al. (2024) propose globally and locally adaptive diffusion (GLAD), a novel diffusion model designed to address the sample reconstruction challenge in unsupervised anomaly detection. GLAD mitigates uneven reconstruction difficulty across different anomalies by predicting specific denoising steps for each sample and spatially adaptive feature fusion, enhancing reconstruction quality and anomaly detection accuracy. Additionally, GLAD introduces an anomaly-oriented training paradigm, incorporating synthetic anomaly samples to improve the reconstruction of abnormal regions, enabling high-performance anomaly detection and localization. To overcome over-generalization and loss of detail in traditional two-stage anomaly detection architectures, Fučka et al. (2024) propose TransFusion, which employs a single-stage iterative process to simultaneously reconstruct and localize anomalies. Yan et al. (2024) propose AnomalySD, a novel anomaly detection framework designed for industrial manufacturing, where normal samples are scarce. AnomalySD leverages the zero-/few-shot repair capability of the stable diffusion model, integrating hierarchical text prompts and a foreground mask mechanism to fine-tune stable diffusion for anomaly detection. During inference, a multi-scale masking strategy and a prototype-guided masking strategy are employed to accurately mask and repair abnormal regions, enabling efficient anomaly detection and localization. Zhang et al. (2024) propose RealNet, a feature selection network designed to enhance industrial anomaly detection by generating realistic and diverse synthetic anomaly samples. RealNet integrates three key modules: (1) a strength-controllable diffusion anomaly synthesis (SDAS) module, which enables precise manipulation of defect location, intensity, and structure, facilitating the simulation of rare, subtle, or complex anomalies; (2) an anomaly-aware feature selection module that selects a representative subset of pre-trained features to enhance anomaly sensitivity; and (3) a reconstruction residual selection module that adaptively selects residuals to improve localization accuracy. Diffusion-based anomaly generation in RealNet effectively addresses the scarcity of defective samples while preserving high visual fidelity and structural diversity.

2.3.5 Generative pre-trained transformer

As illustrated in Fig. 9, a GPT-based framework for industrial visual anomaly detection is presented. Li et al. (2023) propose Myriad, a large-scale multimodal model designed to address the challenge of providing detailed anomaly descriptions and accurate judgments in industrial anomaly detection. Myriad leverages a pre-trained industrial anomaly detection model as a vision expert and employs the vision expert tokenizer to embed anomaly graphs into tokens interpretable by large language models. Additionally, it incorporates an expert-driven vision-language extraction module to generate domain-specific vision-language tokens, enabling precise anomaly detection and comprehensive anomaly descriptions. Zhang et al. (2024) investigate the application of GPT-4V, a large-scale multimodal

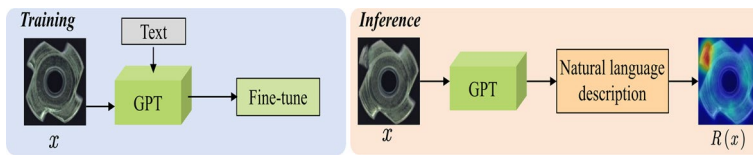


Fig. 9 GPT method. **Training phase:** The input image x and its corresponding textual description are fed into GPT, which is then fine-tuned to generate natural language outputs that align with the image content. Through iterative training, the model progressively learns and captures the semantic features of normal images. **Inference phase:** During inference, given a new input image x , GPT produces a corresponding natural language description based on its learned knowledge. If the generated description significantly deviates from the previously learned semantics of normal samples, it is deemed anomalous

model based on the visual question paradigm, for zero-shot anomaly detection. Zhang et al. propose GPT-4V-AD, a framework that leverages granular region segmentation, carefully crafted prompts, and text-to-segmentation conversion to precisely localize and assess abnormal regions in images. Zhaopeng et al. (2024) propose AnomalyGPT, a novel industrial anomaly detection method based on large-scale vision-language models. Traditional industrial anomaly detection methods require manual threshold setting and struggle to adapt to complex industrial scenarios. AnomalyGPT addresses this limitation by simulating abnormal image data through an image decoder and a prompt learner. AnomalyGPT enables automatic anomaly detection and localization, directly evaluates anomalies without manual threshold adjustment, and supports multi-turn dialogue and few-shot contextual learning, significantly enhancing both the practicality and accuracy of industrial anomaly detection tasks. Xu et al. (2024) propose Customizable-VLM, a method for adapting a general vision-language model to enhance the generalization ability of anomaly detection in industrial scenarios. The approach integrates expert domain knowledge through a multimodal prompting strategy, guiding the model in anomaly detection and reasoning. By unifying the representation of multimodal inputs, the study demonstrates that combining visual and linguistic prompts significantly improves anomaly detection performance across different data modalities while enabling general anomaly detection and reasoning in diverse industrial scenarios. Zhang et al. (2024) propose LogiCode, a novel framework designed to address the challenge of detecting logic anomalies in industrial environments, where traditional methods often struggle. Unlike conventional approaches, LogiCode leverages the logical reasoning capabilities of large language models to identify anomalies such as incorrect component counts or missing elements by automatically generating Python code, while also providing detailed anomaly explanations. LogiCode automates the entire process from logical rule definition to anomaly detection through three key modules: code prompt, code generation, and code execution, offering robust support for industrial quality control and reducing manual inspection workloads. Hattori et al. (2024) propose segment element-based anomaly detection (SEAD), an anomaly detection method designed to address the limitations of existing approaches in detecting logical anomalies that affect entire images. SEAD builds on the traditional ComAD method and detects anomalies through SegGPT segmentation, a memory bank that stores the color and pixel count of each element in normal images, and anomaly detection based on the Euclidean distance between test images and the memory bank.

2.3.6 Segment anything model

Figure 10 presents the SAM-based industrial vision anomaly detection framework. Cao et al. (2023) address the reliance of existing anomaly segmentation models on domain-specific fine-tuning and their limited generalization ability. They propose SAA+, a novel framework for segmenting anomalies in images without training. SAA+ leverages hybrid prompting regularization to adapt the base model using domain expert knowledge and the target image context, enhancing its generalization across different domains. It generates anomalous regions by cascading a prompt-guided object detection and segmentation model and introduces hybrid prompting to reduce linguistic ambiguity and improve segmentation accuracy. Liu et al. (2024) propose UCAD, an unsupervised continual anomaly detection framework designed to address the continual learning challenge in industrial manufacturing caused by the lack of labeled data. UCAD enables unsupervised anomaly detection with continual learning capabilities through contrastively learned prompts, mitigating catastrophic forgetting and reducing computational burden. The framework comprises two core modules: continuous prompt model (CPM) and structure-based contrastive learning (SCL). CPM utilizes a key prompt knowledge memory to guide anomaly predictions while maintaining a consistent task objective. SCL, integrated with the segment anything model, enhances feature representation via contrastive learning, improving anomaly segmentation performance. Li et al. (2024) propose SAM-guided two-stream lightweight model (STLM) enhance model efficiency and mobility in industrial anomaly detection. The framework leverages SAM's strong generalization ability to achieve unsupervised anomaly detection using two lightweight image encoders (two-stream lightweight modules). One stream generates discriminative row-wise and universal feature representations of normal and abnormal regions, while the other reconstructs anomaly-free images, enhancing feature discriminability when detecting anomalies. Additionally, a shared mask decoder and feature aggregation module generate anomaly maps, significantly reducing model parameters and inference time while maintaining high performance, effectively addressing the dual challenges of efficiency and mobility in industrial anomaly detection.

2.4 3D anomaly detection

3D anomaly detection relies on inputs such as depth maps, point clouds, or multi-view projections to identify structural defects. Its perception mode, feature encoding, and model structure are all different from 2D methods that mainly use color or texture. Therefore, this subsection reviews representative 3D visual anomaly detection methods in recent years

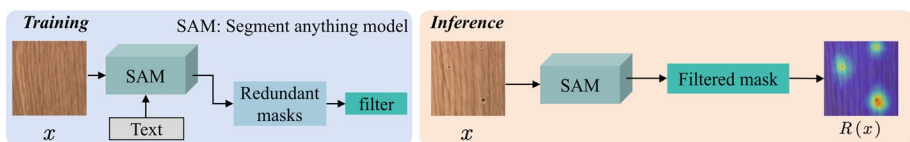


Fig. 10 SAM method. **Training phase:** Given an image x and a textual prompt, the SAM generates multiple candidate segmentation masks. These redundant masks are then filtered to remove extraneous or inaccurate regions, yielding a more precise segmentation result that is stored in the model. **Inference phase:** For a new input image x , SAM again produces segmentation masks, which are further refined based on the previously learned filtering strategy. The resulting filtered masks help localize potential anomalies in the image and facilitate the generation of an anomaly heatmap

to more clearly reflect the technological progress in industrial visual anomaly detection. Specifically, Chen et al. (2023) propose EasyNet, a 3D industrial anomaly detection model designed to address challenges in deployment, storage, and real-time performance faced by existing methods. EasyNet reconstructs segmentation maps of abnormal regions using a multi-scale, multi-modal feature encoder-decoder and employs an attention-based information entropy fusion module to integrate features during inference. EasyNet enables efficient, real-time 3D anomaly detection without pre-trained models or memory banks, significantly enhancing detection performance and speed. Zhu et al. (2024) propose Group3AD, a novel network architecture designed to address challenges in high-resolution 3D power anomaly detection, including a large number of sample points, feature degradation, and the small proportion of anomalous regions. Group3AD enhances anomaly detection efficiency and accuracy through group-level feature comparison and learning. The framework introduces an inter-cluster uniformity network (IUN) and an intra-cluster alignment network (IAN) to optimize feature space distribution. Additionally, it employs an adaptive group center selection (AGCS) mechanism based on geometric information to improve the detection of potentially anomalous regions, significantly enhancing detection accuracy and efficiency. Li et al. (2024) propose DRAIN, a 3D industrial anomaly detection method designed to address the limitations of traditional approaches in effectively utilizing depth information. DRAIN employs a dual reconstruction network, consisting of a 3D reconstruction network and a 3D segmentation network, to process RGB and depth images using an encoder-decoder architecture. Additionally, it integrates a lightweight multi-layer perceptron (MLP) and a dual attention mechanism to fuse features, enabling high-performance 3D industrial anomaly detection. Zhou et al. (2024) propose R3D-AD, a novel anomaly detection method designed to address the challenges of high resource consumption and incomplete anomaly region reconstruction in existing 3D anomaly detection approaches. R3D-AD reconstructs point clouds by incorporating a diffusion model and employs Patch-Gen, a self-supervised learning strategy that simulates diverse defect shapes to enhance model generalization. Ultimately, R3D-AD achieves complete reconstruction of anomalous point clouds and efficient anomaly detection. Cao et al. (2024) propose complementary pseudo-multimodal features (CPMF) to address the challenge of point cloud anomaly detection. CPMF enhances detection performance by fusing local geometric and global semantic features. CPMF combines local geometric features extracted via 3D hand-crafted descriptors with global semantic features derived from multi-view images using a 2D pre-trained neural network. Additionally, CPMF employs multi-view image projection and feature aggregation techniques to further improve anomaly detection accuracy. Cheng et al. (2024) propose MVP-PCLIP, a multi-view projection framework (MVP) that projects point cloud data into multi-view depth images and integrates anomaly detection with pre-trained vision-language models (VLMs). MVP-PCLIP addresses the challenges of poor generalization, high data acquisition costs, and the "cold start" problem in traditional unsupervised point cloud anomaly detection. Additionally, MVP-PCLIP enhances model performance by introducing learning visual cues and adaptive text prompts, which improve the model's focus on anomalous regions and the accuracy of textual descriptions, significantly boosting zero-shot point cloud anomaly detection. Zhou et al. (2024) propose PointAD, a novel zero-shot 3D anomaly detection method designed to detect anomalies in unseen objects without target 3D training samples. PointAD constructs a unified framework that integrates R-CNN, 3D point clouds, and 2D pixel representations to enhance 3D anomaly understanding and detection. PointAD

employs multi-view rendering to project 3D point clouds into multiple 2D views, extracts pixel representations using a CLIP model, and maps them back to 3D space. Additionally, hybrid representation learning optimizes text prompt comprehension from both 3D and 2D information, enabling robust anomaly detection across diverse 3D objects. Furthermore, PointAD can seamlessly incorporate 2D RGB information during inference for multimodal 3D anomaly detection without requiring additional training. Liu et al. (2024) address the scarcity of high-precision point cloud anomaly detection datasets and the absence of systematic benchmarks by constructing Real3D-AD, a comprehensive dataset for 3D industrial anomaly detection. Additionally, they propose Reg3D-AD, a registration-based 3D anomaly detection method that leverages a feature memory bank to store local and global features, significantly enhancing detection accuracy. Costanzino et al. (2024) present CFM, a novel multimodal industrial anomaly detection framework designed to overcome the limitations of traditional methods in identifying complex industrial anomalies. CFM extracts 2D image and 3D point cloud features using a Transformer-based architecture and trains a lightweight neural network to learn the feature mapping between modalities. Anomalies are detected by identifying inconsistencies between observed and mapped features. Additionally, CFM introduces a layer pruning technique to optimize memory and computational efficiency, significantly enhancing detection and segmentation performance while achieving faster inference speeds and lower memory usage.

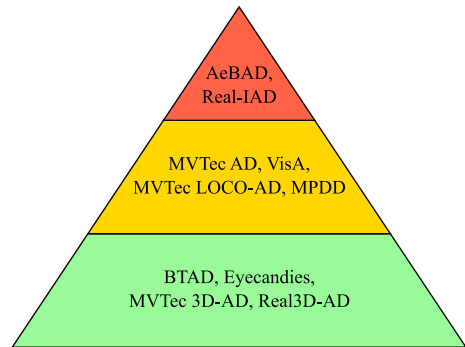
3 Datasets

In industrial visual inspection tasks, anomalies can generally be divided into two categories: structural anomalies and semantic anomalies. Structural anomalies are usually manifested as geometric defects or physical damage on the surface of the product, such as scratches, cracks, holes, dents, etc., and can be effectively identified by local texture or edge information. Semantic anomalies are more biased towards errors at the high-level semantic level, such as missing parts, logical mismatches, and unreasonable positions, which are often difficult to directly determine using low-level pixel features. In the existing data sets, MVTec AD mainly contains structural anomalies, covering typical types of industrial surface defects, such as fabric (fabric tears), bottle (deformed bottle mouth), transistor (soldering point anomaly), etc. MVTec LOCO AD further introduces anomalies at the semantic level, such as “logical anomaly” (incorrect component quantity, misalignment during installation, etc.), which is closer to the complex visual judgment requirements in actual industrial scenarios. Therefore, we have specially marked the performance differences of each method in dealing with different types of anomalies in the experimental comparison to help understand the boundaries of the methods’ capabilities.

This section presents a comprehensive overview of widely used industrial anomaly detection datasets. Specifically, Sect. 3.1 discusses 2D datasets, while Sect. 3.2 focuses on 3D datasets. In addition, this paper provides a multi-dimensional analysis and categorization of these datasets. Based on factors such as task complexity, visual structural diversity, cross-domain generalization, and alignment with real-world industrial scenarios, the datasets are organized into three hierarchical tiers, as illustrated in Fig. 11.

- The top tier comprises the Real-IAD and AeBAD datasets, which are characterized

Fig. 11 Hierarchical classification of industrial anomaly detection benchmark datasets. The pyramid represents increasing complexity and real-world alignment from bottom to top, based on criteria including visual diversity, defect realism, annotation quality, and domain variability



by high-resolution imagery, multiple viewpoints, cross-domain variability, and video-based data. These datasets are highly representative in terms of defect diversity, scene realism, and task complexity.

- The middle tier includes MVTec AD, VisA, MVTec LOCO-AD, and MPDD, all of which provide high-quality annotations and support a broad range of anomaly detection tasks. Owing to their versatility and reliability, these datasets are widely employed for benchmarking, method validation, and performance assessment.
- The bottom tier consists of BTAD, Eyecandies, MVTec 3D-AD, and Real3D-AD, which are primarily suited for specialized subtasks such as geometric structure modeling. Nevertheless, certain limitations exist within these datasets, particularly concerning annotation granularity and domain generalizability.

3.1 2D anomaly detection

The MVTec AD dataset Bergmann et al. (2019) models real-world industrial production scenarios and serves as a benchmark for anomaly detection. It consists of fifteen industrial product categories, including five textured and ten object categories. The training set contains only normal images, whereas the test set includes both normal and anomalous images with pixel-level annotations for the latter. As a result, this dataset has become a widely used benchmark for unsupervised anomaly detection and localization.

The VisA dataset Zou et al. (2022) employs a self-supervised learning approach, pretraining the model using a Spot-the-Difference task to improve anomaly detection and segmentation performance. The dataset comprises 12 subsets, each representing a distinct object, with a total of 10,821 images—9,621 normal and 1,200 anomalous. Four subsets feature printed circuit boards (PCB1, PCB2, PCB3, and PCB4) with complex structures, including transistors, capacitors, and chips. For multi-instance objects in the same field of view, VisA includes four subsets: Capsules, Candles, Macaroni1, and Macaroni2, with Capsules and Macaroni2 exhibiting significant positional and pose variations. Additionally, VisA includes four subsets—Cashew, Chewing Gum, Fryum, and Pipe Fryum—in which objects are approximately aligned. The anomalous images encompass various defects, which fall into two categories: surface defects (e.g., scratches, dents, stains, or cracks) and structural defects (e.g., misalignment or missing parts).

The BTAD dataset Mishra et al. (2021) features defects on both the body and surface of industrial products, comprising 1,800 normal images for training. In comparison to MVTec

AD, BTAD does not provide annotations for fine details and is better suited for large-scale anomaly detection.

Unlike previous datasets such as MVTec AD and VisA, which primarily focus on defect detection, AeBAD Zhang et al. (2023) emphasizes diversity within the same data category across different domains. AeBAD aims to automatically detect anomalies in aeroengine blades to ensure operational stability. It consists of two subsets: the single-blade dataset (AeBAD-S) and the video-based anomaly detection dataset (AeBAD-V). AeBAD-S comprises single-blade images with varying scales and aspect ratios. Its primary characteristics include misalignment and significant scale variations. Additionally, there is a domain shift between the normal images in the training and test sets, primarily due to variations in lighting and viewpoints. AeBAD-V comprises videos of assembled aeroengine blades, which facilitate anomaly detection during blisk rotation. A key challenge in AeBAD-V is the discrepancy in viewing angles between the training and test sets.

MPDD Jezek et al. (2021) is an unsupervised dataset for industrial anomaly detection, designed to assess a model's performance in detecting and segmenting defects in industrial production. Specifically, MPDD focuses on painted metal parts and consists of 6,600 images spanning six categories. The training set comprises 5,280 images, while the test set comprises 1,320 images. MPDD features multiple objects across different orientations, positions, and distances, under diverse lighting conditions and non-uniform backgrounds. Additionally, it provides pixel-level annotations for anomalous images, facilitating precise defect localization.

MVTec LOCO-AD Bergmann et al. (2022) is a benchmark dataset for assessing unsupervised anomaly detection and localization algorithms. It comprises 3,644 images captured from five real-world industrial inspection scenarios, incorporating both structural and logical anomalies. Structural anomalies encompass scratches, dents, and contamination on manufactured products. Logical anomalies result from implicit constraint violations, e.g., misplaced allowed objects or missing required objects.

3.2 3D Anomaly Detection

MVTec 3D-AD Bergmann et al. (2022) is a large-scale 3D benchmark dataset for unsupervised anomaly detection and localization. It features high-resolution scans acquired using industrial-grade 3D sensors and consists of ten object categories, each containing defect-free training and validation images, along with test images exhibiting various defects such as scratches, dents, holes, contamination, and deformation. Additionally, MVTec 3D-AD provides detailed pixel-wise annotations for each anomalous test image.

Eyecandies Bonfiglioli et al. (2022) is a benchmark dataset for unsupervised anomaly detection and localization. It comprises 14,234 images across ten candy categories, each posing unique challenges, including complex textures, self-occlusion, and specular reflections.

Real3D-AD Liu et al. (2024) is the first industrial anomaly detection dataset incorporating point cloud data. It contains 1,254 images across 12 categories. Real3D-AD integrates RGB images for surface feature extraction and 3D point cloud data for capturing geometric structures to identify object shapes and surface anomalies. Additionally, it includes depth images, which provide a more detailed representation of object shapes using depth information. All objects and parts in Real3D-AD are sourced from real-world industrial production

and exhibit various defects, including scratches, holes, cracks, deformations, and missing parts.

Real-IAD Wang et al. (2024) is a large-scale multi-view industrial anomaly detection dataset. It comprises over 150,000 high-resolution images of industrial products made from 30 different materials, including metal, plastic, wood, ceramics, and mixed materials. Each product is captured from five distinct perspectives, offering multi-view coverage with detailed pixel-level defect annotations. Real-IAD not only surpasses existing datasets in scale but also introduces image-level evaluation metrics and, for the first time, a fully supervised industrial anomaly detection setting. This better simulates real-world industrial anomaly detection scenarios, advancing both research and practical applications in anomaly detection technology.

4 Evaluation metrics and results

4.1 Evaluation metrics

In industrial anomaly detection, precision quantifies a model's ability to correctly identify anomalies. Mathematically, it is defined as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (50)$$

where TP (True Positive) refers to correctly predicted anomalies, while FP (False Positive) refers to instances where normal samples are incorrectly classified as anomalies. A higher precision indicates better model performance. Recall quantifies the proportion of true anomalies that are correctly identified by the model. Mathematically, it is defined as:

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (51)$$

where FN (False Negative) refers to an actual anomaly that is incorrectly classified as normal. A higher recall indicates that the model can correctly identify more anomalous cases. For a balanced evaluation, the $F1_{\text{score}}$ serves as a harmonic mean when both precision and recall are given equal importance.

$$F1_{\text{score}} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (52)$$

A higher $F1_{\text{score}}$ indicates better model performance in balancing precision and recall.

Image-AUROC is derived from the image-level Area Under the Curve (AUC) of the ROC curve (Receiver Operating Characteristic Curve), which measures a model's performance in image-level classification. For each image, the model outputs an anomaly score representing the probability of the image being anomalous. Based on the true labels (normal/abnormal) and the predicted scores for all images, an ROC curve is plotted, with the False Positive Rate (FPR) on the horizontal axis and the True Positive Rate (TPR) on the

vertical axis. The AUC value is obtained by calculating the area under the ROC curve. The AUROC value ranges from 0 to 1, with higher values indicating better model classification performance. It primarily evaluates a model's ability to distinguish between normal and abnormal images. Additionally, it assesses the presence of anomalies in an entire image without considering their specific locations.

Pixel-AUROC is derived from pixel-level ROC curves, measuring a model's ability to distinguish anomalous pixels from normal ones. For each pixel, the model outputs an anomaly score representing the probability of the pixel being anomalous. The true labels (normal/abnormal) of all pixels are compared with their corresponding predicted scores, and the ROC curve is then constructed using the pixel-level false positive rate (FPR) on the horizontal axis and the pixel-level true positive rate (TPR) on the vertical axis. The area under the ROC curve is computed to obtain the AUC value.

Per-Region Overlap (PRO) is a metric designed for pixel-level anomaly segmentation. It quantifies the overlap between the predicted and ground truth defect regions. Unlike traditional approaches that merely count pixel-level TP and FP, PRO first segments the ground truth defect areas into one or more independent regions and then computes their overlap with the corresponding predicted regions. This approach better captures the overall effectiveness of defect detection and localization, instead of evaluating correctness at an individual pixel level.

sPRO Bergmann et al. (2022) builds upon PRO with a normalization strategy to provide a fairer evaluation of anomaly detection performance. By normalizing PRO, sPRO mitigates evaluation bias arising from the presence of extremely small or large defect regions, thus ensuring a more equitable comparison across different algorithms.

In industrial anomaly detection, it is sometimes necessary to account for variations in anomaly shape, size, and unknown detection scenarios. To address this challenge, AUPRO is commonly employed. AUPRO is derived from the area under the PRO curve and quantifies a model's overall performance in anomaly region detection, with a particular emphasis on localization accuracy. For each anomalous region, the IoU between the ground truth and predicted anomaly regions is computed as a function of varying prediction thresholds. The horizontal axis represents the Normalized Pixel Coverage (NPC) of the predicted region, defined as the ratio of the predicted region's area to the ground truth anomaly region's area. The vertical axis represents the IoU values, and the area under the PRO curve is integrated to obtain the AUPRO value. Compared with AUROC, AUPRO places greater emphasis on the accuracy of anomaly localization instead of solely evaluating classification performance. Model sensitivity to varying anomaly sizes can be analyzed by adjusting the prediction threshold.

Overall, Image-AUPRO, Pixel-AUPRO, PRO, AUPRO, and sPRO form a multi-level evaluation framework for industrial defect detection, particularly in pixel-level segmentation tasks. They emphasize precise localization and identification of defects, incorporating various calculation methods and threshold adjustments to address extreme cases. In industrial applications requiring precise localization and comprehensive detection, these metrics offer a more targeted and interpretable evaluation compared to pixel-level precision and recall alone, thereby enabling researchers to more effectively compare and refine various defect detection algorithms.

4.2 Experiment results

This section presents the experimental results of industrial anomaly detection, aiming to provide researchers with a clear comparative analysis in order to more accurately select anomaly detection methods suitable for different scenarios. All experimental results are excerpted from the original papers of each method, fully adopting their publicly available training–testing splits, input resolutions, backbone networks, and evaluation scripts. To ensure a representative and diverse evaluation, we cover the main categories introduced in Sect. 2. Specifically, these include reconstruction-based (e.g., CutPaste, PatchCore), feature embedding-based (e.g., FastFlow, SPADE), and diffusion-based models (e.g., RealNet). The selection also spans different supervision paradigms, including unsupervised, weakly supervised, and few-shot approaches. This broad coverage allows for a more comprehensive comparison across various learning strategies and detection paradigms. When exploring the applicability in different manufacturing environments, the quality control method proposed by Bono et al. (2023) adopts a strategy that combines soft sensors with anomaly detectors, which effectively solves the problem of quality monitoring under highly unstable working conditions. Based on this research, the experimental data presented in this paper were collected from diverse industrial production environments, which fully reflects the actual performance of the anomaly detection method in each scenario. As shown in Table 2, we summarize the highlights of some methods in order to provide researchers with a strong reference when selecting anomaly detection strategies. Table 3 summarizes the experimental results of different methods and their resource consumption on the MVTec AD and VisA datasets. Tables 4 and 5 illustrate the image-level and pixel-level results for each category on the MVTec AD dataset, respectively. Similarly, Tables 6 and 7 show the image-level and pixel-level results for each category on the VisA dataset. Table 8 presents the experimental results on the MVTec 3D AD dataset, and Table 10 shows the experimental results on the MVTec LOCO AD dataset. In addition, Fig. 12 presents the experimental results on the MVTec AD dataset, intuitively illustrating the differences in anomaly detection performance among various methods through visualizations. Figure 13 presents the visualization results of anomaly detection experiments on the VisA dataset. These results clearly highlight each method's capability in detecting anomalies in complex structures and multi-instance objects, as well as their effectiveness on single-instance objects. Figure 14 presents the visualized experimental results on the MVTec 3D AD dataset, clearly demonstrating the performance of different methods in handling three-dimensional anomaly detection tasks. The MVTec 3D AD dataset provides high-quality RGB-D data with realistic 3D surface defects in multiple object categories. Unlike 2D datasets that focus primarily on texture anomalies, this dataset enables the evaluation of methods capable of capturing geometric deformations, depth anomalies, and irregular surface topology, which are becoming increasingly important in modern industrial inspection scenarios. Furthermore, MVTec 3D AD has been widely used in recent literature as a benchmark for evaluating 3D vision anomaly detection algorithms, ensuring the comparability and reproducibility of experimental results. These characteristics make it a suitable and representative dataset for validating the performance of reconstruction- and depth-aware anomaly detection methods. Figure 15 demonstrates the detection results on the MVTec LOCO AD dataset, and Fig. 16 shows the detection results on the BTAD, MPDD, AeBAD-S datasets which compares the advantages of the methods in a more intuitive way.

Table 2 Highlights of some industrial anomaly detection methods

Methods	Highlights
AdaCLIP Cao et al. (2024)	1. Hybrid learnable prompt and semantic fusion modules 2. The pre-trained text-image alignment capability of CLIP is effectively leveraged to significantly enhance the performance of anomaly pattern perception and detection
AnomalyCLIP Zhou et al. (2024)	1. Use generic text prompts that are independent of specific objects to abstract the universal features of “normal” and “abnormal” in an image 2. Reduce reliance on specific category knowledge and improve adaptability when generalizing to anomaly detection across different scenarios
AnoVL Deng et al. (2023)	1. The introduction of a training-independent “value-to-value attention mechanism” enables the extraction of key features without requiring additional training 2. The incorporation of test-time adaptation during testing significantly enhances detection and localization performance in zero- and few-shot scenarios
CLIP-AD Chen et al. (2023)	1. Propose an innovative text prompt design paradigm and a staged dual-path model to address the challenge of aligning image features with text features 2. Align and progressively integrate image and text features to significantly enhance the accuracy and stability of anomaly detection
DiAD He et al. (2024)	1. By leveraging semantic guidance and spatial-aware feature fusion, global semantic information is preserved, while anomalous regions are reconstructed with high precision 2. Detection and localization performance is consistently enhanced in multi-class scenarios
FAIR Liu et al. (2023)	1. A frequency-aware image restoration method is employed to effectively differentiate normal and abnormal patterns using high-frequency information 2. A simple UNet can achieve state-of-the-art results in industrial visual anomaly detection while remaining easy to deploy and scale
GLASS Chen et al. (2024)	1. Propose a controllable global–local anomaly synthesis strategy to generate increasingly realistic training samples for weak defects 2. It excels in detecting weak defects and enhances sensitivity to small anomalies while maintaining both high accuracy and efficiency
WinCLIP Jeong et al. (2023)	1. Adopts a windowed feature extraction strategy combined with language-guided anomaly definition to enable fine-grained analysis of local regions 2. Exhibits excellent performance in industrial visual scenarios, effectively balancing local detail detection and overall semantic understanding

Table 3 Performance comparison and computational resource requirements of different methods on MVTec AD and VisA datasets

Method	Reference	Learning strategy	Backbone	Image AUROC	Pixel AUROC	AUPRO	FPS	Batch size	GPU mem-ory(≈ G)
DestSeg Zhang et al. (2023)	KBS2024	Supervised	WRN50-2	98.85/–	98.55/–	–/–	74.7	32	10.4
DRAEM Zavrtanik et al. (2021)	ICCV2021	Weakly supervised	U-Net	98.05/79.48	97.49/91.37	–/–	54	8	11.9
WinCLIP Jeong et al. (2023)	CVPR2023	Weakly supervised	ViT-B-16-plus-240	79.87/72.51	87.55/91.41	72.52/69.93	–	256	13.1
REB Shuai et al. (2024)	KBS2024	Weakly supervised	ResNet18	99.20/–	–/–	–/–	–	96	16.2
FAIR Liu et al. (2023)	arXiv2023	Weakly supervised	U-Net	98.10/96.73	98.21/97.69	93.87/–	–	8	7.3
PUAD Sugawara and Imamura (2024)	arXiv2024	Weakly supervised	R101-2	98.30/–	–/–	–/–	–	1	1.2
AdaCLIP Cao et al. (2024)	ECCV2024	Weakly supervised	ViT-L-14-336	86.69/34.01	88.50/57.09	–/–	–	1	11.0
AnomalyCLIP Zhou et al. (2024)	ICLR2023	Weakly supervised	ViT-L-14-336	91.56/81.81	91.07/95.33	96.34/85.14	–	8	7.3
AnoVL Deng et al. (2023)	arXiv2023	Weakly supervised	ViT-B-16-plus-240	92.49/79.20	90.85/89.89	77.69/71.12	–	8	3.2
DiAD He et al. (2024)	AAAI2024	Unsupervised	ResNet50	96.13/86.76	96.75/96.04	90.89/–	–	2	22.5
GLASS Chen et al. (2024)	ECCV2024	Unsupervised	WRN50-2	99.58/98.06	98.85/98.34	92.02/85.78	–	8	5.1
HGAD Yao et al. (2024)	ECCV2024	Unsupervised	EfficientNet-B6	98.90/96.80	98.10/98.80	–/–	–	8	7.4
SimpleNet Liu et al. (2023)	CVPR2023	Unsupervised	WRN50-2	99.52/87.18	97.31/96.83	–/–	77	8	6.6
MMR Zhang et al. (2023)	COMIND2023	Unsupervised	ResNet50	98.35/–	97.20/–	92.61/–	–	16	5.8
PADIM Defard et al. (2021)	ICPR2021	Unsupervised	WRN50-2	95.50/–	97.33/–	–/–	60.32	32	0.5
NSA Schlitter et al. (2022)	ECCV2022	Unsupervised	ResNet18	97.20/–	96.06/–	–/–	–	64	6.5
RIAD Zavrtanik et al. (2021)	PR2021	Unsupervised	WRN50-2	91.65/–	94.15/–	–/–	–	4	10.6
CFA Sungwook et al. (2022)	IEEE2022	Unsupervised	WRN50-2	99.17/–	98.15/–	94.44/–	–	4	4.8
UniAD You et al. (2022)	NeurIPS2022	Unsupervised	EfficientNet-B4	96.5/88.78	96.8/98.32	–/–	56.8	8	3.8
RD Deng and Li (2022)	CVPR2022	Unsupervised	WRN50-2	98.46/–	97.81/–	93.93/–	90.6	32	14.1
RD++ Tien et al. (2023)	CVPR2023	Unsupervised	WRN50-2	99.44/–	98.25/–	94.99/–	≈ 134	16	12.7
PatchCore Roth et al. (2022)	CVPR2023	Unsupervised	WRN50-2	99.10/–	98.06/–	93.41/–	22.3	2	5.5
CFLOW-AD Gudovskiy et al. (2022)	WACV2022	Unsupervised	WRN50-2	97.87/–	97.35/–	94.60/–	27	32	4.8
MuSc Zhou et al. (2024)	ICLR2024	Unsupervised	WRN50-2	97.72/92.53	97.11/98.71	–/–	47.6	1	4.5

In practical industrial visual anomaly detection, parameter selection and tuning remain critical challenges due to the diversity of inspection environments, object types, and defect patterns. Different industrial scenarios may require different settings for thresholds, feature dimensions, and backbone depths to achieve optimal performance. To address these issues, several strategies have been commonly adopted. One approach is to initialize parameters using configurations that perform robustly across multiple public datasets (e.g., MVTec AD, VisA), providing a good starting point. Additionally, lightweight tuning can be performed on a small set of validation images collected from the target environment to calibrate decision thresholds or refine feature selection layers. Furthermore, the use of adaptive modules—such as attention mechanisms, multi-scale fusion, or learnable normalization layers—has been proven effective in reducing the sensitivity of models to fixed parameter values. These techniques help models generalize better under changing lighting conditions, background textures, and defect shapes. As industrial systems move toward real-world deployment, balancing generality and adaptability in parameter design remains an essential consideration for anomaly detection frameworks. These parameter adaptation strategies have been explored and validated in several representative works, including CutPaste Li et al. (2021), PatchCore Roth et al. (2022), and WinCLIP Jeong et al. (2023), which demonstrate strong robustness and transferability across industrial datasets.

The experimental results further reveal the differences between different algorithms in dealing with structural and semantic anomalies. Taking MVTec LOCO AD as an example, we have calculated the AUROC metrics of multiple methods under structural anomalies (SA) and logical/semantic anomalies (LA). As can be seen from Table 9, GCAD performs better in dealing with structural anomalies (e.g., the SA-AUROC of the Connector class reaches 99.0), while ComAD has certain advantages in logical anomaly detection, indicating that the method is more capable of modeling contextual semantic relationships. These results suggest that different types of anomalies may require different modeling strategies. Structural anomalies rely more on fine-grained boundary reconstruction and low-level feature modeling, while semantic anomalies rely more on global context understanding and high-level feature comparison.

Based on the results presented in Tables 3, 4, 5, 6, 7, 8, and 9 and Figures 12, 13, 14, 15 and 16, we observe that feature embedding-based methods such as PatchCore and FastFlow perform well in detecting structural anomalies, showing strong generalization across datasets with clear defect patterns. Diffusion-based methods like RealNet achieve superior performance in both localization and anomaly synthesis, particularly in complex backgrounds or low-contrast regions. Meanwhile, reconstruction-based methods demonstrate solid detection capabilities but tend to be more sensitive to anomaly scale and texture variability. These findings highlight the trade-offs between detection accuracy, robustness, and computational efficiency, providing useful guidance for selecting appropriate methods under different industrial scenarios.

4.3 Parameter selection

In real-world deployment of industrial visual anomaly detection (IVAD) systems, the selection and optimization of hyperparameters remains one of the major challenges in model implementation. This difficulty primarily stems from the diversity and complexity of industrial environments, where production lines can vary significantly in terms of camera angles,

Table 4 Image-level anomaly detection performance (AU-ROC%) comparison of methods for the anomaly detection on the MVTecAD dataset

Method	Texture		Object										Mean			
	Carpet	Grid	Leather	Tile	Wood	Bottle	Cable	Capsule	Hazelnut	Metal_nut	Pill	Screw		Toothbrush	Transistor	Zipper
SimpleNet Liu et al. (2023)	99.40	99.25	100.0	99.96	100.0	100.0	99.93	97.69	99.82	99.95	98.88	98.28	99.72	100.0	99.97	99.52
MMMR Zhang et al. (2023)	99.50	100.0	100.0	98.70	99.10	100.0	97.70	97.10	100.0	99.90	98.10	92.30	100.0	95.10	97.70	98.35
PADIM	98.40	89.80	98.80	95.90	99.00	99.60	85.50	87.00	84.10	97.40	86.90	74.50	94.70	92.50	74.10	90.55
Defard et al. (2021)	99.90	95.70	100.0	97.40	98.80	99.80	92.20	91.50	93.30	99.20	94.40	84.40	97.20	97.80	90.90	95.50
DRAEM Zavr- tanik et al. (2021)	96.27	100.0	100.0	100.0	99.21	96.90	93.40	96.05	100.0	99.36	96.75	99.26	99.72	94.17	99.68	98.05
SMCC Yongh- eng et al. (2023)	100.0	99.30	100.0	100.0	99.60	100.0	96.60	96.60	100.0	99.60	96.50	91.10	100.0	100.0	98.50	98.52
NSA Schlüter et al. (2022)	95.60	99.90	99.90	100.0	97.50	97.70	94.50	95.20	94.70	98.70	99.20	90.20	100.0	95.10	99.80	97.20
RIAD Zavr- tanik et al. (2021)	84.20	99.60	100.0	98.70	93.00	99.90	81.90	88.40	83.30	88.50	83.80	84.50	100.0	90.90	98.10	91.65
CFA Sung- wook et al. (2022)	97.30	99.20	100.0	99.40	99.70	100.0	99.80	97.30	100.0	100.0	97.90	97.30	100.0	100.0	99.60	99.17

Table 4 (continued)

Method	Texture		Object										Mean				
	Carpet		Grid	Leather	Tile	Wood	Bottle	Cable	Capsule	Hazelnut	Metal_nut	Pill		Screw	Toothbrush	Transistor	Zipper
RD Deng and Li (2022)	98.90		100.0	100.0	99.30	99.20	100.0	95.00	96.30	99.90	100.0	96.60	97.00	99.50	96.70	98.50	98.46
RD++ Tien et al. (2023)	100.0		100.0	100.0	99.70	99.30	100.0	99.20	99.00	100.0	100.0	98.40	98.90	100.0	98.50	98.60	99.44
PatchCore Roth et al. (2022)	98.70		98.20	100.0	98.70	99.20	100.0	99.50	98.10	100.0	100.0	96.60	98.10	100.0	100.0	99.40	99.10
CFLOW-AD Gudovskiy et al. (2022)	99.23		96.89	99.61	97.71	94.49	98.76	97.64	98.98	98.82	98.56	98.95	98.10	98.56	93.28	98.41	97.87
MuSc Zhou et al. (2024)	99.88		98.75	100.0	100.0	98.51	99.92	98.88	95.89	99.64	96.92	95.83	82.74	100.0	99.42	99.42	97.72
REB Shuai et al. (2024)	99.60		100.0	100.0	100.0	99.70	100.0	100.0	99.30	99.70	100.0	99.00	94.70	96.10	99.90	100.0	99.20
DestSeg Zhang et al. (2023)	99.60		100.0	100.0	100.0	99.30	100.0	98.13	97.33	98.54	98.83	96.78	96.33	99.44	98.50	100.0	98.85
WinCLIP Jeong et al. (2023)	96.39		57.89	99.15	94.70	95.79	99.29	57.61	61.19	92.68	60.90	83.50	58.95	79.17	63.96	96.95	79.87
HGAD Yao et al. (2024)	100.0		100.0	100.0	100.0	99.60	100.0	97.00	98.90	99.50	100.0	99.00	98.00	93.10	98.10	99.80	98.87

Table 4 (continued)

Method	Texture			Object											Mean	
	Carpet	Grid	Leather	Tile	Wood	Bottle	Cable	Capsule	Hazelnut	Metal_nut	Pill	Screw	Toothbrush	Transistor		Zipper
FAIR Liu et al. (2023)	99.72	98.83	99.97	100.0	99.56	100.0	97.23	95.17	99.18	97.90	99.37	90.45	100.0	97.46	96.72	98.10
DiAD He et al. (2024)	98.31	98.21	100.0	95.86	96.61	100.0	90.00	94.22	97.10	96.13	95.07	91.19	96.77	96.30	96.23	96.13
AdaCLIP Cao et al. (2024)	99.88	92.86	99.97	99.49	97.94	97.70	81.69	84.48	59.61	55.79	87.89	84.77	94.58	78.04	85.67	86.69
Anom- alyCLIP Zhou et al. (2024)	100.0	97.80	99.80	100.0	96.90	88.70	70.30	89.50	97.20	92.40	81.10	82.10	85.30	93.90	98.40	91.56
AnoVL Deng et al. (2023)	99.30	98.10	100.0	100.0	99.30	96.30	87.80	79.10	94.00	97.40	86.60	78.00	92.20	86.40	92.90	92.49

The best results are emphasized in **bold**

Table 5 Pixel-level anomaly detection performance (AU-ROC%) comparison of methods for the anomaly detection on the MVTec AD dataset

Method	Texture			Object												Mean
	Carpet	Grid	Leather	Tile	Wood	Bottle	Cable	Capsule	Hazelnut	Metal_nut	Pill	Screw	Toothbrush	Transistor	Zipper	
SimpleNet Liu et al. (2023)	97.28	95.83	99.16	92.34	93.77	98.00	97.44	98.88	97.58	98.69	98.27	99.14	98.45	96.73	98.09	97.31
MMMR Zhang et al. (2023)	98.75	99.00	99.22	95.63	94.80	98.30	95.43	98.03	98.50	95.94	98.42	99.47	98.35	90.18	97.99	97.20
PADIM	98.80	93.60	99.00	91.70	94.00	98.10	94.90	98.20	97.90	96.70	94.60	97.20	98.60	96.80	97.60	96.51
Defard et al. (2021)	99.00	96.50	98.90	93.90	94.10	98.20	96.80	98.60	97.90	97.10	96.10	98.30	98.70	97.50	98.40	97.33
DRAEM Zavr- tanik et al. (2021)	96.23	99.56	98.88	99.50	97.17	99.27	95.43	94.04	99.55	98.73	97.62	99.68	98.12	89.97	98.63	97.49
SMCC Yongh- eng et al. (2023)	99.10	97.70	99.30	96.20	94.50	98.80	98.60	99.10	98.90	99.30	98.80	98.30	99.10	97.70	98.90	98.29
NSA Schlüter et al. (2022)	95.50	99.20	99.50	99.30	90.70	98.30	96.00	94.30	97.60	98.40	98.50	96.50	94.90	88.00	94.20	96.06
RIAD Zavr- tanik et al. (2021)	96.30	98.80	99.40	89.10	85.80	98.40	84.20	92.80	96.10	92.50	95.70	98.80	98.90	87.70	97.80	94.15
CFA Sung- wook et al. (2022)	99.28	98.12	99.37	95.21	91.53	98.84	98.97	99.11	98.85	99.15	98.93	98.91	98.96	98.06	99.02	98.15

Table 5 (continued)

Method	Texture			Object												Mean
	Carpet	Grid	Leather	Tile	Wood	Bottle	Cable	Capsule	Hazelnut	Metal_nut	Pill	Screw	Toothbrush	Transistor	Zipper	
RD Deng and Li (2022)	98.90	99.30	99.40	95.60	95.30	98.70	97.40	98.70	98.90	97.30	98.20	99.60	99.10	92.50	98.20	97.81
RD++ Tien et al. (2023)	99.20	99.30	99.40	96.60	95.80	98.80	98.40	98.80	99.20	98.10	98.30	99.70	99.10	94.30	98.80	98.25
PatchCore Roth et al. (2022)	99.00	98.70	99.30	95.40	95.00	98.60	98.40	98.80	98.70	98.40	97.40	99.40	98.70	96.30	98.80	98.06
CFLOW-AD Gudovskiy et al. (2022)	99.20	96.26	99.60	97.53	94.15	98.05	97.02	98.77	98.73	98.22	98.96	97.09	98.50	90.39	97.76	97.35
MuSc Zhou et al. (2024)	99.45	98.18	99.72	97.89	97.25	98.47	95.74	98.95	99.38	86.02	97.49	98.78	99.53	91.31	98.42	97.11
DestSeg Zhang et al. (2023)	98.93	98.57	99.80	99.27	98.22	99.23	97.28	99.00	99.62	97.79	99.03	99.42	99.45	93.39	99.18	98.55
WinCLIP Jeong et al. (2023)	98.66	79.42	98.56	86.57	92.38	91.14	54.09	90.85	98.15	71.09	94.48	94.35	97.24	68.96	97.36	87.55
HGAD Yao et al. (2024)	99.40	99.10	99.60	97.00	96.40	98.40	96.20	99.00	98.70	97.20	98.50	99.40	98.70	95.40	98.80	98.12
FAIR Liu et al. (2023)	99.62	99.37	99.65	98.39	97.01	98.37	98.26	93.99	99.48	98.28	98.28	98.85	99.16	95.09	99.35	98.21

Table 5 (continued)

Method	Texture			Object												Mean
	Carpet	Grid	Leather	Tile	Wood	Bottle	Cable	Capsule	Hazelnut	Metal_nut	Pill	Screw	Toothbrush	Transistor	Zipper	
DiAD He et al. (2024)	98.37	98.19	99.11	92.97	93.52	98.35	94.97	96.97	98.22	97.54	95.64	98.05	99.02	94.03	96.27	96.75
AdaCLIP Cao et al. (2024)	98.99	97.21	99.32	87.54	95.53	86.52	69.43	93.53	95.62	72.03	87.26	97.78	92.93	57.84	95.93	88.50
Anom- alyCLIP Zhou et al. (2024)	98.80	97.30	98.60	94.70	96.40	90.40	78.90	95.80	97.20	74.60	91.80	97.50	91.90	70.80	91.30	91.07
AnoVL Deng et al. (2023)	98.70	96.40	98.90	96.20	95.70	93.10	82.20	90.70	95.40	76.70	82.90	93.00	93.20	77.90	91.70	90.85

The best results are emphasized in **bold**

Table 6 Image-level anomaly detection performance (AU-ROC%) comparison of methods for the anomaly detection on the VisA dataset

Method	Complex Structure				Multiple Instances				Single Instances				Mean
	Peb1	Peb2	Peb3	Peb4	Macaroni1	Macaroni2	Capsules	Candle	Cashew	Chewing gum	Fryum	Pipe_fryum	
DRAEM Zavtitanik et al. (2021)	71.90	78.50	76.60	97.40	69.80	59.40	83.40	69.30	81.70	93.70	89.20	82.80	79.48
UniAD You et al. (2022)	94.20	91.10	82.20	99.00	82.80	76.00	70.30	95.80	94.30	97.50	86.90	95.30	88.78
SimpleNet Liu et al. (2023)	91.60	92.40	89.10	97.00	85.90	68.30	74.10	84.10	88.00	96.40	88.40	90.80	87.18
ViTAD Zhang et al. (2025)	95.80	90.60	90.90	99.10	85.80	79.10	79.20	90.40	87.80	94.90	94.30	97.80	90.48
FAIR Liu et al. (2023)	97.70	97.20	99.30	98.70	97.90	91.70	93.90	96.00	96.50	96.20	96.80	98.80	96.73
MuSe Zhou et al. (2024)	89.10	93.61	93.16	98.34	89.45	68.34	88.27	96.27	98.56	98.26	98.54	98.44	92.53
DiAD He et al. (2024)	88.10	91.40	86.20	99.60	85.70	62.50	58.20	92.80	91.50	99.10	89.80	96.20	86.76
WinCLIP Jeong et al. (2023)	78.61	57.02	51.58	88.56	60.54	59.60	65.70	85.03	71.04	87.56	71.70	93.12	72.51
AdaCLIP Cao et al. (2024)	25.47	34.56	46.17	15.50	41.01	44.62	55.82	37.75	32.91	22.44	30.71	21.21	34.01
AnomalyCLIP Zhou et al. (2024)	83.70	60.10	60.70	94.40	87.80	73.30	87.50	81.10	71.70	97.40	91.10	92.90	81.81
AnoVL Deng et al. (2023)	82.90	50.40	69.70	84.70	68.40	65.00	80.30	96.00	89.60	95.60	87.50	80.30	79.20

The best results are emphasized in **bold**

lighting conditions, defect manifestations, and dimensional tolerances of acceptable products. Such variations hinder the unified optimization of key hyperparameters, including decision thresholds, feature dimensions, backbone network depth, and batch size. Moreover, environmental changes during production-such as equipment wear, illumination drift, or material variation-can introduce domain shifts, leading to performance degradation over time. These dynamic factors further complicate parameter selection and reduce the long-term reliability of fixed hyperparameter configurations.

To address these issues, a common practice is to adopt parameter configurations that have shown robust performance across multiple public datasets (e.g., MVTec AD, VisA) as initial baselines. These are then lightly fine-tuned using small-scale in-field data to calibrate thresholds or adjust feature-related settings. Additionally, incorporating adaptive modules-such as attention mechanisms, multi-scale feature fusion, and learnable normalization layers-can mitigate the model's sensitivity to fixed parameters and improve generalization under shifting production conditions. These strategies have been widely validated across various industrial scenarios, demonstrating their effectiveness and practical utility in improving deployment robustness.

In the actual deployment of industrial vision anomaly detection systems, threshold setting plays a critical role in determining both practicality and economic efficiency. An overly low threshold can result in excessive false alarms, leading to frequent production line interruptions and increased manual inspection costs. Conversely, a high threshold may cause real defects to be missed, posing significant quality risks. Therefore, selecting an appropriate threshold is essential for effective model deployment. A common approach involves using a small amount of labeled data in the offline phase to determine the initial threshold by maximizing the F1 score. When only normal samples are available, the threshold is typically set at a high-percentile point of the anomaly score distribution that meets a target false alarm rate, based on statistical modeling of normal data. However, once the model is deployed, the anomaly score distribution often drifts due to factors such as lighting variations, equipment aging, and batch-to-batch differences in production. To cope with this, sliding window statistics or cost-sensitive dynamic calibration methods are employed to maintain an optimal balance between false positives and false negatives. Industrial practice has shown that a well-designed threshold management strategy can reduce the false positive rate by over 40% while maintaining the false negative rate below 0.1%, underscoring the critical role and practical value of threshold optimization in real-world applications.

5 Challenge and future directions

This section outlines several key future prospects that aim to assist the development of industrial anomaly detection field.

5.1 Multimodal and multi-view dataset

In industrial anomaly detection, a single RGB image is often insufficient to fully capture the anomaly characteristics of a product or component. Many potential anomalies are not limited to the surface of an object. Internal defects such as cracks and cavities rely more on penetrative imaging techniques (e.g., X-rays and ultrasound) for detection. In contrast,

Table 7 Pixel-level anomaly detection performance (AU-ROC%) comparison of methods for the anomaly detection on the VisA dataset

Method	Complex Structure				Multiple Instances				Single Instances				Mean
	Pcb1	Pcb2	Pcb3	Pcb4	Macaroni1	Macaroni2	Capsules	Candle	Cashew	Chewing gum	Fryum	Pipe_fryum	
DRAEM Zavtaniik et al. (2021)	94.70	92.30	90.80	94.40	95.00	94.60	97.10	82.20	80.70	91.10	92.40	91.10	91.37
UniAD You et al. (2022)	99.20	98.00	98.20	97.20	99.00	97.30	97.40	99.00	99.00	99.10	97.30	99.10	98.32
SimpleNet Liu et al. (2023)	99.20	96.60	97.20	93.90	98.90	93.20	97.10	97.60	98.90	97.90	93.00	98.50	96.83
ViTAD Zhang et al. (2025)	99.50	97.90	98.20	99.10	98.50	98.10	98.20	96.20	98.50	97.80	97.50	99.50	98.25
FAIR Liu et al. (2023)	99.70	98.90	99.30	97.80	99.70	91.70	93.90	98.00	98.80	98.30	96.80	99.40	97.69
MuSe Zhou et al. (2024)	99.49	97.39	98.05	98.69	99.51	97.16	98.73	99.36	99.33	99.54	97.93	99.39	98.71
DiAD He et al. (2024)	98.70	95.20	96.70	97.00	94.10	93.60	97.30	97.30	90.90	94.70	97.60	99.40	96.04
WinCLIP Jeong et al. (2023)	88.71	87.52	88.53	95.50	92.80	85.09	87.78	91.15	92.17	97.92	91.46	98.32	91.41
AdaCLIP Cao et al. (2024)	73.16	61.96	71.36	31.91	80.08	69.47	61.24	10.29	69.52	30.23	72.06	53.85	57.09
AnomalyCLIP Zhou et al. (2024)	94.50	92.50	88.00	95.80	98.10	97.30	95.10	98.70	92.50	99.10	94.10	98.30	95.33
AnoVL Deng et al. (2023)	94.10	87.70	84.10	95.60	88.10	85.50	84.70	96.00	90.10	96.50	89.00	87.30	89.89

The best results are emphasized in **bold**

point cloud data, while providing 3D shape information, cannot penetrate deep into objects. Therefore, relying solely on visible light or surface-based detection techniques may still result in missing key internal anomalies. To improve detection coverage and accuracy, it is essential to construct a comprehensive dataset incorporating multimodal data (e.g., RGB images, point clouds, X-ray imaging, and ultrasonic scanning). Through multimodal collaboration, industrial systems can gain a more comprehensive evaluation of the internal and external states of products, thus enabling more precise localization and identification of complex anomalies. This presents both a challenge in data collection and fusion and a key future direction in industrial anomaly detection. Additionally, multi-perspective data collection solutions are indispensable for capturing object anomalies from multiple viewpoints. Taking Real-IAD as an example, this dataset is designed for real industrial scenarios and employs large-scale, multi-perspective collection methods. Each object is captured from at least five different viewpoints to fully account for potential appearance variations. This type of multi-view and multimodal data resource not only adapts to varying object forms and complex backgrounds in industrial scenarios but also supports deep learning algorithms in defect recognition, localization, and classification, ultimately forming a solid foundation for developing a high-precision and robust industrial anomaly detection system.

5.2 Precision and latency in real-time

In the industrial visual inspection scenario, there are significant differences between real-time anomaly detection and batch processing detection methods in terms of system design, performance requirements and application goals. Real-time detection emphasizes millisecond image acquisition and inference response, and is widely used in scenarios that are extremely sensitive to response speed, such as high-takt production lines, semiconductor defect detection. Take EfficientAD as an example. It can achieve a single-image inference latency of about *2ms* on the NVIDIA RTX A6000 GPU, processing up to 600 images per second, demonstrating excellent real-time detection performance and providing a reference for industrial deployment. However, in actual deployment, deep learning models still face many computational bottlenecks in achieving real-time performance. First, models such as convolutional neural networks and Transformers have a large number of parameters, which require a large number of matrix operations and convolution operations, resulting in a significant increase in inference latency. Second, industrial images are often high-resolution, and a large number of intermediate features and model parameters can cause memory bottlenecks and bandwidth pressure. At the same time, data pre-processing and transmission also pose challenges for I/O performance. The limited computing power of edge devices often makes it difficult to support the stable operation of complex models. Power consumption and thermal design constraints are also issues that cannot be ignored. In addition, the latency of data collection, pre-processing, model inference and post-processing has a cumulative effect. Without system-level coordinated optimization, it significantly slows down the overall response. In contrast, the batch processing method is more suitable for offline quality analysis, defect statistics, and exception tracing tasks. This method allows for greater freedom in model expressiveness by caching or centrally processing images, which allows for the use of more complex model structures and more powerful post-processing modules. However, there is a feedback delay in the detection results, which is difficult to meet real-time industrial scenarios that require immediate intervention or control.

Table 8 Performance comparison and computational resource requirements of different methods on MVTec 3D AD dataset

Method	Reference	Backbone	Image AUROC	Pixel AUROC	AUPRO	Batch size	GPU memory($\approx$ G)
3D-ADS Horwitz and Hoshen (2023)	CVPRW2023	WideResNet50	87.80	99.19	95.97	4	22.4
MMR Zhang et al. (2023)	COMIND2023	ResNet50	98.00	99.00	95.23	64	5.8
CPMF Cao et al. (2024)	PR2024	ResNet18	85.37	87.85	83.17	1	2.6
HGAD Yao et al. (2024)	ECCV2024	EfficientNet-B6	98.00	99.00	95.23	8	7.3
CFM Costanzino et al. (2024)	CVPR2024	ViT-B/8(DINO)	98.00	99.00	95.23	4	18.7

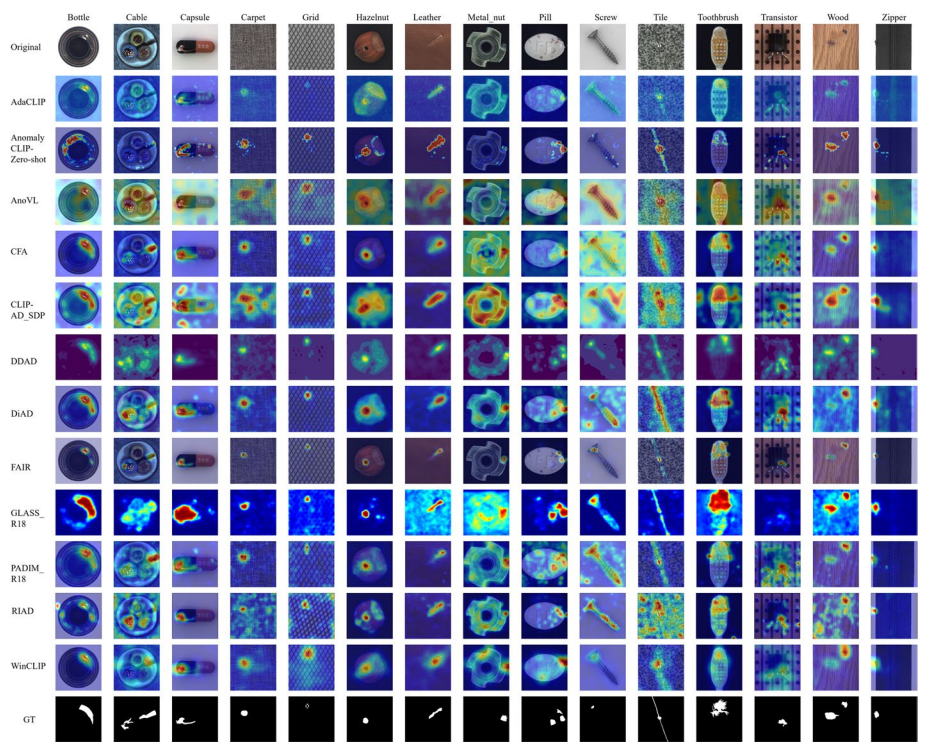


Fig. 12 Experimental results in MVTec AD dataset



Fig. 13 Experimental results in VisA dataset

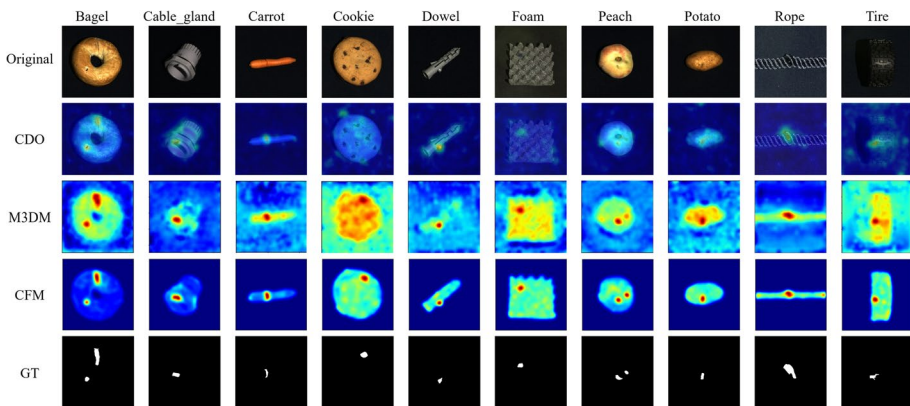


Fig. 14 Experimental results in MVTec 3D AD dataset

From a technical implementation perspective, real-time detection usually relies on lightweight models, model compression techniques, and efficient inference engines, and is deployed on edge devices, FPGAs, or AI modules. Batch processing, on the other hand, can leverage cloud resources to run large models, enabling higher resolution or multi-modal detection capabilities. In summary, real-time detection emphasizes low latency, high responsiveness, and edge deployment capabilities. Batch processing methods, on the other hand, focus on complexity and depth of analysis, and are more suitable for strategic quality

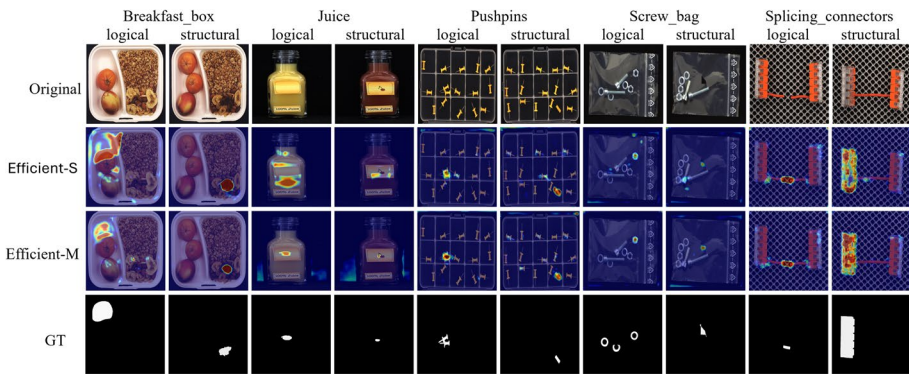


Fig. 15 Experimental results in MVTEC LOCO AD dataset

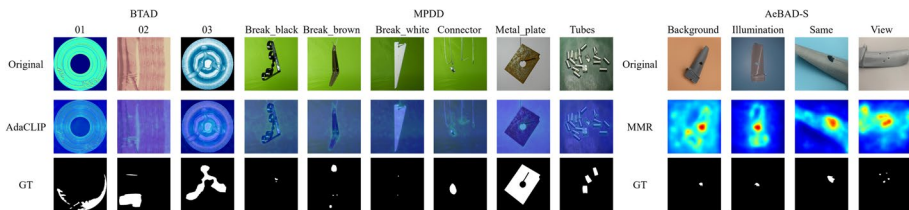


Fig. 16 Experimental results in BTAD, MPDD, and AeBAD-S dataset

monitoring and post-model evaluation. In actual industrial deployment, a hybrid architecture that combines real-time detection at the edge with batch processing and analysis in the cloud has become the mainstream trend, taking into account both immediate intervention and system scalability.

5.3 Generative anomaly detection

In the field of industrial visual anomaly detection, generative detection methods are gaining increasing prominence. These methods learn the distribution of 'normal' data through deep generative models, enabling anomaly detection during inference. This approach is especially effective in industrial scenarios where normal samples are readily available, but abnormal samples are rare or difficult to obtain. As generative models continue to advance in network architectures, training strategies, inference efficiency, and interpretability, industrial visual anomaly detection is evolving towards greater accuracy, enhanced adaptability, and improved interpretability. Notably, when combined with weakly supervised, semi-supervised, and multimodal data fusion techniques, the potential of generative anomaly detection in industrial applications can be further unlocked, enabling highly automated, reliable industrial systems and more robust solutions to adapt to dynamic manufacturing processes and diverse defect patterns.

Table 10 Image-level logical and structural anomaly detection performance (LA/SA AUROC) on the MVTec LOCO AD dataset

Category	Breakfast_box	Juice_bottle	Pushpins	Screw_bag	Splicing_connectors	Average	Mean
DRAEM Zavrtan- ik et al. (2021)	75.10/85.40	97.80/90.80	55.70/81.50	56.20/85.00	75.20/95.50	72.00/87.64	79.82
AST Rudolph et al. (2023)	80.00/79.90	91.60/95.50	65.10/77.80	80.10/95.90	81.80/89.40	79.72/87.70	83.71
Patch- Core Roth et al. (2022)	74.80/80.10	70.90/98.50	76.60/87.90	67.80/92.00	79.20/88.00	73.86/89.30	81.58
ComAD Liu et al. (2023)	94.50 /68.51	90.46/80.24	79.65/66.56	88.90 /93.90	84.10/63.56	87.52 /74.55	81.04
RD Deng and Li (2022)	66.70/60.30	93.60/95.20	63.60/84.80	54.10/89.20	75.30/ 95.90	70.66/85.08	77.87
GCAD Berg- mann et al. (2022)	87.00/80.90	100.0/98.90	97.50 /74.90	56.00/70.50	89.70 /78.30	86.04/80.70	83.37

The best results are shown in **bold**

5.4 Lightweighting and hardware acceleration

With the rapid advancement of industrial automation and intelligent manufacturing, the application value of the industrial internet of things and edge computing technology in industrial visual anomaly detection is becoming increasingly prominent. Industrial internet of things connects high-resolution industrial cameras, intelligent sensors, and actuators to continuously and frequently collect and distribute multi-source image data from the production process, providing rich and real-time visual information for anomaly detection models. However, traditional cloud-based detection methods are limited by image transmission delays, bandwidth bottlenecks, and data privacy issues, making it difficult to meet the actual needs of high-takt production lines for low latency and high responsiveness.

To address these challenges, edge computing is gradually becoming the mainstream choice. By deploying the visual anomaly detection model directly on edge devices close to the data source, the system can achieve local inference and millisecond-level response, significantly reducing transmission latency and communication load. In this process, lightweight neural network architectures are optimized using techniques such as pruning, quantization, and knowledge distillation, enabling stable operation on resource-constrained hardware platforms. For example, edge GPUs: the NVIDIA Jetson Orin Nano runs an INT8-quantised EfficientAD model at approximately 1.5ms per 640×640 frame, fulfilling the 600fps requirement for high-takt production lines. Edge TPUs: the Google Coral TPU deliv-

Table 9 Image-AUROC and AUPRO result on the MVTEC 3D-AD dataset

Method	Bagel	Cable gland	Carrot	Cookie	Dowel	Foam	Peach	Potato	Rope	Tire	Mean
Depth											
Depth GAN Bergmann et al. (2022)	53.0/11.1	37.6/7.2	60.7/21.2	60.3/17.4	49.7/16.0	48.4/12.8	59.5/0.3	48.9/4.2	53.6/44.6	52.1/7.5	52.4/14.2
Depth AE Bergmann et al. (2022)	46.8/14.7	73.1/6.9	49.7/29.3	67.3/21.7	53.4/20.7	41.7/18.1	48.5/16.4	54.9/6.6	56.4/54.5	54.6/14.2	54.6/20.3
Depth VM Bergmann et al. (2022)	51.0/28.0	54.2/37.4	46.9/24.3	57.6/52.6	60.9/48.5	69.9/31.4	45.0/19.9	41.9/38.8	66.8/54.3	52.0/38.5	54.6/37.4
Voxel GAN Bergmann et al. (2022)	38.3/44.0	62.3/45.3	47.4/87.5	63.9/75.5	56.4/78.2	40.9/37.8	61.7/39.2	42.7/63.9	66.3/77.5	57.7/38.9	53.8/58.8
Voxel AE Bergmann et al. (2022)	69.3/26.0	42.5/34.1	51.5/58.1	79.0/35.1	49.4/50.2	55.8/23.4	53.7/35.1	48.4/65.8	63.9/1.5	58.3/18.5	57.2/34.8
Voxel VM Bergmann et al. (2022)	75.0/45.3	74.7/34.3	61.3/52.1	73.8/69.7	82.3/68.0	69.3/28.4	67.9/34.9	65.2/63.4	60.9/61.6	69.0/34.6	69.9/49.2
3D-ST Bergmann and Sattlegger (2023)	86.2/–	48.4/–	83.2/–	89.4/–	84.8/–	66.3/–	76.3/–	68.7/–	95.8/–	48.6/–	74.8/–
FPFH Zavrtnik et al. (2024)	82.5/97.3	55.1/87.9	95.2/98.2	79.7/90.6	88.3/89.2	58.2/73.5	75.8/97.7	88.9/98.2	92.9/95.6	65.3/96.1	78.2/92.4
AST Rudolph et al. (2023)	88.1/–	57.6/–	96.5/–	95.7/–	67.9/–	79.7/–	99.0/–	91.5/–	95.6/–	61.1/–	83.3/–
M3DM Wang et al. (2023)	94.1/94.3	65.1/81.8	96.5/97.7	96.9/88.2	90.5/88.1	76.0/74.3	88.0/95.8	97.4/97.4	92.6/95.0	76.5/92.9	87.4/90.6
EasyNet Chen et al. (2023)	73.5/16.0	67.8/3.0	74.7/68.0	86.4/75.9	71.9/75.8	71.6/6.9	71.3/22.5	72.5/73.4	88.5/79.7	68.7/50.9	74.7/47.2
DRAIN Li et al. (2024)	81.3/76.8	72.0/87.2	93.6/97.8	92.4/89.5	82.9/86.9	72.0/56.8	87.6/86.2	85.9/85.4	90.4/92.8	69.2/75.2	82.7/83.5

Table 9 (continued)

Method	Bagel	Cable gland	Carrot	Cookie	Dowel	Foam	Peach	Potato	Rope	Tire	Mean
RGB											
DifferNet Rudolph et al. (2021)	85.9/-	70.3/-	64.3/-	43.5/-	79.7/-	79.0/-	78.7/-	64.3/-	71.5/-	59.0/-	69.6/-
PADiM Defard et al. (2021)	97.5/-	77.5/-	69.8/-	58.2/-	95.9/-	66.3/-	85.8/-	53.5/-	83.2/-	76.0/-	76.4/-
PatchCore Roth et al. (2022)	87.6/90.1	88.0/94.9	79.1/92.8	68.2/87.7	91.2/89.2	70.1/56.3	69.5/90.4	61.8/93.2	84.1/90.8	70.2/90.6	77.0/87.6
STFPM Wang et al. (2021)	93.0/-	84.7/-	89.0/-	57.5/-	94.7/-	76.6/-	71.0/-	59.8/-	96.5/-	70.1/-	79.3/-
CFLOW-AD Gudovskiy et al. (2022)	94.1/85.5	93.0/91.9	82.7/95.8	79.5/86.7	99.0/96.9	88.6/50.0	73.1/88.9	47.1/93.5	98.6/90.4	74.5/91.9	83.0/87.2
RD4AD Deng and Li (2022)	97.5/-	98.7/-	94.3/-	57.5/-	99.9/-	83.0/-	86.3/-	61.8/-	98.4/-	89.9/-	86.7/-
AST Rudolph et al. (2023)	94.7/-	92.8/-	85.1/-	82.5/-	98.1/-	95.1/-	89.5/-	61.3/-	99.2/-	82.1/-	88.0/-
M3DM Wang et al. (2023)	94.4/95.2	91.8/ 97.2	89.6/ 97.3	74.9/89.1	95.9/93.2	76.7/ 84.3	91.9/97.0	64.8/ 95.6	93.8/ 96.8	76.7/ 96.6	85.1/ 94.2
EasyNet Chen et al. (2023)	98.2/75.1	99.2/82.5	91.7/91.6	95.3/59.9	91.9/69.8	92.3/69.9	84.0/91.7	78.5/82.7	98.6/88.7	74.2/63.6	90.4/77.6
DRAIN Li et al. (2024)	99.4/97.4	99.5/95.2	93.2/93.8	98.5/90.1	97.5/ 97.3	93.6/67.7	84.7/92.2	79.4/83.4	99.8/90.9	82.6/78.2	92.8/88.6

Table 9 (continued)

Method	Bagel	Cable gland	Carrot	Cookie	Dowel	Foam	Peach	Potato	Rope	Tire	Mean
RGB+depth (2022)	53.8/42.1	37.2/42.2	58.0/77.8	60.3/69.6	43.0/49.4	53.4/25.2	64.2/28.5	60.1/36.2	44.3/40.2	57.7/ 63.1	53.2/47.4
Depth AE Bergmann et al. (2022)	64.8/43.2	50.2/15.8	65.0/80.8	48.8/49.1	80.5/84.1	52.2/40.6	71.2/26.2	52.9/21.6	54.0/71.6	55.2/47.8	59.5/48.1
Depth VM Bergmann et al. (2022)	51.3/38.8	55.1/32.1	47.7/19.4	58.1/57.0	61.7/40.8	71.6/28.2	45.0/24.4	42.1/34.9	59.8/26.8	62.3/33.1	55.5/33.6
Voxel GAN Bergmann et al. (2022)	68.0/66.4	32.4/62.0	56.5/76.6	39.9/74.0	49.7/78.3	48.2/33.2	56.6/58.2	57.9/79.0	60.1/63.3	48.2/48.3	51.8/63.9
Voxel AE Bergmann et al. (2022)	51.0/46.7	54.0/75.0	38.4/80.8	69.3/55.0	44.6/76.5	63.2/47.3	55.0/72.1	49.4/91.8	72.1/11.9	41.3/17.0	53.8/56.4
Voxel VM Bergmann et al. (2022)	55.3/51.0	77.2/33.1	48.4/41.3	70.1/71.5	75.1/68.0	57.8/27.9	48.0/30.0	46.6/50.7	68.9/61.1	61.1/36.6	60.9/47.1
3D-ST Bergmann and Sattlegger (2023)	-/95.0	-/48.3	-/98.6	-/92.1	-/90.5	-/63.2	-/94.5	-/98.8	-/97.6	-/54.2	-/83.3
FPFH Zavrtanik et al. (2024)	91.8/-	74.8/-	96.7/-	88.3/-	93.2/-	58.2/-	89.6/-	91.2/-	92.1/-	88.6/-	86.5/-
AST Rudolph et al. (2023)	98.3/-	87.3/-	97.6/-	97.1/-	93.2/-	88.5/-	97.4/-	98.1/-	100.0/-	79.7/-	93.7/-
M3DM Wang et al. (2023)	99.4/-	90.9/-	97.2/-	97.6/-	96.0/-	94.2/-	97.3/-	89.9/-	97.2/-	85.0/-	94.5/-
EasyNet Chen et al. (2023)	99.1/83.9	99.8/86.4	91.8/95.1	96.8/61.8	94.5/82.8	94.5/83.6	90.5/94.2	80.7/88.9	99.4/91.1	79.3/52.8	92.6/82.1
DRAIN Li et al. (2024)	99.5/97.9	99.7/86.3	95.2/97.7	97.2/ 93.2	94.3/91.6	95.3/85.4	96.3/95.1	82.3/96.3	99.0/92.4	84.1/56.5	94.3/89.2

The best results are marked in bold

ers an inference latency of 12ms on a MobileNet-based detector after full INT8 compilation. Industrial FPGAs: AMD (Xilinx) Kria KV260 and Versal AI Edge modules, pre-loaded with the DPU or Vitis-AI, achieve 5–7ms per 1080p frame, while guaranteeing deterministic latency and low power consumption. In addition, dedicated hardware accelerators (such as TPU and FPGA) offer unique advantages in terms of parallelism, energy efficiency and customizability, providing strong support for visual inspection tasks deployed at the edge. Furthermore, an edge-cloud collaborative architecture is also widely used in industrial scenarios: real-time detection and initial screening are performed at the edge, while the cloud is responsible for complex scene modeling and high-precision model updates, thereby balancing response speed and detection accuracy. In the future, industrial visual inspection systems continue to integrate low-power vision modules, multimodal sensing fusion mechanisms, and adaptive structural search technologies at the edge to better meet the multiple needs of industrial sites for high accuracy, high efficiency, and low power consumption, and promote the widespread implementation of intelligent quality control systems.

In addition, while improving system efficiency and deployment flexibility, the network security of the model is also an issue that cannot be ignored in industrial deployment. AI-driven anomaly detection models may face various risks such as adversarial attacks (misleading the model by constructing inputs), data poisoning (tampering with training samples), and model leakage (reconstructing input data through output). If these security risks are exploited, they not only affect system reliability, but may also lead to abnormal production lines or even the disclosure of sensitive data. Therefore, security mechanisms such as adversarial training, access control, and encrypted model compression should be introduced in actual deployments to ensure the robustness and trustworthiness of industrial intelligent detection systems.

5.5 Incremental learning for industrial visual anomaly detection

In industrial production, evolving product designs, manufacturing processes, and defect types often render static models ineffective, necessitating costly full retraining. Incremental learning offers a practical solution by enabling models to adapt to emerging defect categories and shifting data distributions while retaining previously acquired knowledge. This approach significantly reduces the time and computational resources required for frequent retraining and supports sustained model performance. In real-world industrial visual anomaly detection deployments, incremental learning typically unfolds in two complementary stages. The first stage involves the automatic discovery of novel anomalies, commonly achieved through clustering or visual inspection of flagged samples. Once a new defect pattern is confirmed, the model undergoes targeted incremental updates—either during low-load periods or directly on edge devices—to swiftly integrate the new characteristics. The second stage focuses on gradual adaptation to distributional shifts caused by changing lighting conditions, material variations, or equipment wear. In this phase, the model is continuously updated using small batches of incoming data, often leveraging techniques such as experience replay or regularization constraints to prevent catastrophic forgetting of previously learned defect information. Looking ahead, as production lines become increasingly autonomous and flexible, incremental learning will play a vital role in ensuring that industrial visual anomaly detection systems maintain high robustness and responsiveness

to unforeseen defect modes. This capability is essential for sustaining consistent quality control in dynamic industrial environments.

5.6 Federated learning and privacy-preserving techniques

With the increasing emphasis on data security in industrial settings, federated learning and privacy-preserving techniques have garnered growing attention in visual anomaly detection. Traditional training paradigms often rely on centralized data aggregation, which poses potential risks of data leakage and regulatory non-compliance. In contrast, federated learning enhances data security and legal compliance by distributing model training across multiple local clients and performing parameter updates without transmitting raw image data.

In industrial visual inspection tasks, data distributions often vary significantly across different factories, devices, or production lines. Federated learning facilitates collaborative model training while preserving the domain-specific characteristics of each node, thereby improving generalization in heterogeneous environments. Moreover, the integration of differential privacy, homomorphic encryption, and secure aggregation techniques can further prevent the inference or reconstruction of sensitive data from model updates, ensuring robust protection at the edge.

Recent studies on federated learning-based anomaly detection frameworks have begun to explore strategies for knowledge sharing among distributed edge nodes, especially in response to concept drift and non-stationary data streams. Looking forward, the integration of federated learning with multimodal fusion, model compression, and knowledge distillation is expected to drive the development of more secure, efficient, and scalable industrial visual anomaly detection systems.

6 Conclusion

In this paper, we systematically explore the current developments and emerging trends in industrial visual anomaly detection, spanning methodologies to practical applications. Based on supervision levels, we classify existing methods into five categories—fully supervised, semi-supervised, self-supervised, weakly supervised, and unsupervised—based on their supervisory signals. We also conduct an in-depth analysis of their design motivations, implementation challenges, and performance characteristics with specific case studies. Additionally, we review embedding-based and reconstruction-based industrial visual anomaly detection methods and compare their strengths and limitations in enhancing detection accuracy and reducing inference costs. Through a comprehensive analysis of existing datasets and their application scenarios, we summarize the main challenges in industrial visual anomaly detection, including insufficient data diversity, high annotation costs, and the pressing need for lightweight models. To address these challenges, we propose several promising research directions, including introducing generative models, developing lightweight methods, fusing multimodal data, and developing systematic and integrated approaches for industrial anomaly detection. We hope that our in-depth exploration of these directions further drive the real-world adoption of industrial visual anomaly detection, enhancing efficiency and reliability in industrial production and quality control.

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Data availability Data will be made available on request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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